

Is this time different? Impact of AI in output, employment and inequality across low, middle and high-income countries

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ABSTRACT

Recent developments in artificial intelligence (AI) have revived concerns about technological unemployment and the increase in inequality due to technological change. Many studies of the impact of automation have considered a static picture of economies that put less emphasis on their potential for job creation through product innovation. They also tend to focus on developed countries and do not consider the potential impact on developing countries through trade and changes in specialisation patterns. This paper studies the impact of AI on the GDP and employment in low, middle and high-income countries, as well as on the income inequality across countries, based on computer simulations of a multi-country, multi-sector macroeconomic model with endogenous technological change proposed by Freire (2017, 2019). It considers the effect of AI in reducing labour requirements and the potential effects of increasing the pace of product and process innovation. The analysis suggests that the introduction of AI, even when assuming large effects of labour substitution, results in only small changes at the aggregated level of GDP and employment. However, there is an increase in income inequality across countries, and there are considerable shifts in jobs between production sectors and R&D. This suggests that AI may not cause mass technological unemployment but would still result in large distributional changes in low, middle and high-income countries.

1. Introduction

There has been a growing concern with technological unemployment and potential increasing inequality caused by rapid technological change and emerging technologies such as artificial intelligence (AI), robotics and automation in general (Autor, 2015; DESA, 2017; Acemoglu and Restrepo, 2018b; *IEEE Spectrum: Technology, Engineering, and Science News*, 2019; United Nations, 2019; UNCTAD, 2019; *The New York Times*, 2020).

However, many studies tend to focus on the effect of technological change destroying jobs and put less emphasis on market mechanisms that counterbalance that process as well as on the opportunities for the creation of new jobs and occupations that result from product innovation.² Moreover, less emphasis has been placed on the study of the impact in developing countries. This impact could be through the adoption of AI and automation in the production structure of developing countries, but also through the adoption of these technologies in developed countries and the effect on employment and growth in

developing countries through trade and changes in export specialization.

The rapid pace of innovation in developed countries, and the introduction of a potentially disruptive technology such as AI, could also affect the opportunities for innovation and catching-up of poorer countries. This has important implications for inequalities between countries given that the vast differences among countries that we observe today began with the start of the first industrial revolution and each technological revolution has been associated with higher disparity among countries (UNCTAD, 2021). Innovation in developing countries is usually carried out through the emulation of production that already exists in more industrialized countries. AI could increase the technological gap between developed and developing countries, reducing the possibilities for successful emulation. Therefore, we need to consider the country's productive capabilities and the possibilities for technology transfer when assessing the effects of rapid technological change.

The research question of the paper is: How can AI impact GDP, employment, and income inequality across low, middle, and high-

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² See Vivarelli (2014) for a survey of the economic literature on the effects of technological change on employment, which includes a comprehensive discussion of the economic theory of the mechanisms that compensate the labour saving effects of innovation.

income countries? Its contribution is to examine, through computer simulations of a model proposed by Freire (2017, 2019), the impact of AI, on output and employment of countries at different levels of income, and the resulting impact on inequalities between countries. It advances the current discussions related to the impact of AI by underscoring the importance of assessing such impact at countries at different levels of income that are interrelated through trade. This paper broadens the scope by exploring the differential effects of AI on countries at varying levels of economic development, providing a global analysis of how AI may widen or reduce economic inequality.

The paper is an application of Freire (2017, 2019) model, with no assumptions modified or extensions to the existing model. Specifically, the paper examines how AI influences labour requirements and the pace of product and process innovation using computer simulations of a multi-country multi-sector model with endogenous technological progress and change in consumption patterns proposed by Freire (2017, 2019). The paper presents an implementation of Freire's model with specific simulations and new outcomes not previously analyzed. In these simulations, I consider the effect of a technology that is introduced in the economy and serves as a proxy for AI in the sense that it reduces the labour required for production to the point of eliminating jobs in sectors in which AI is applied. However, it could also result in lower prices of products, which in turn result in increasing demand, not only for the product that experienced innovation but also other products due to the increase in real income, which creates jobs. The adoption of AI also results in cost savings and higher profits, which in the model are invested in R&D and the potential introduction of new production sectors and the improvement of existing production.

I also consider the effect of these technologies in accelerating the pace of innovation and technological change. This is done through two channels. First, I consider that these technologies complement labour in the R&D sector and increase the rate of process and product innovation. Second, I consider that they increase the possibility of successful combinations with other technologies, which could also increase the rate of innovation.

The changes described above affect the prices of products, which affects the specialization pattern of trading economies. They also affect innovation in these countries and the possibilities for catching up. Here I focus on the effects on output and employment in low, middle, and high-income countries, as well as on inequalities between countries.

The results of the paper suggest that the impact of AI in the aggregated GDP and employment in low, middle and high-income countries could be very marginal. However, it could have a negative effect on income inequality across countries in most scenarios considered, and there is a considerable shift in the distribution of jobs between production and R&D sectors in all countries. The results suggest that, even under the assumption of replacing most of the labour required in the sectors that it is adopted, the introduction of AI would have a similar effect to the introduction of previous innovation paradigms in terms of the international relations of countries, trade specialization and potential for catching up.

The paper is structured into six sections. The next section presents a summary review of the relevant literature. Section three presents a brief description of the model and discusses the setup of the computer

simulations. Section four presents and discusses the results of these simulations. Section five briefly discusses the results in relation to the existing literature on the potential impact of AI on jobs. It also discusses some of the assumptions of the model and how they have affected the results. Section six concludes.

2. Literature review

Concerns about technological unemployment date back to the onset of the industrial revolution (Mokyr et al., 2015; Campa, 2018). Historically, technological change has created more jobs and of different kinds.³ Some recent empirical studies have confirmed the potential for job creation of AI. For example, Damioli et al. (2021) present an analysis based on 3500 companies that patented AI-relevant technologies between 2000 and 2016 and found that companies more focused on AI technologies have a higher impact on job creation. However, most empirical studies on the impact of AI and robots on jobs have suggested a reduction in employment, as highlighted by Barbieri et al. (2019) in an updated review of empirical evidence on the relationship between new technologies and employment. Empirical evidence is essential because economic theory has not provided a clear answer on the effects of innovation on employment (Vivarelli, 2014). Nevertheless, there is a need to expand the theoretical work further to better understand the impact of new technologies on developing countries and through different channels such as trade.⁴

Acemoglu and Restrepo (2017) note that the literature on the impact of AI and automation shows two main lines of research: one based on expert judgment to assess the risk of jobs and tasks being automated, and the other based on economic models and considering compensatory mechanisms to estimate broader economic impact. The literature based on expert judgment resulted in a significant variation in estimates of impact (Arntz et al., 2017; Frey and Osborne, 2017; McKinsey Global Institute, 2017; PwC, 2018; Nedelkoska and Quintini, 2018). A challenge in this stream of literature is that their estimates are related to a risk of automation due to current and expected technological aspects, but it does not take into consideration any economic factor, such as changes in the structure of the economies, in productivity rates and their effects of prices and wages, consumption patterns and their effects on production and employment, and on patterns of international trade and their effects on competitiveness, production, employment and growth.

The other stream of the literature uses economic models to estimate impact considering compensatory mechanisms. Dosi et al. (2019) classify the compensatory mechanisms as classical mechanisms (new machines and the sectoral shift from downstream sectors to upstream machine-producing sectors, decrease in prices in competitive markets triggering increase in demand and employment, and new investments

³ Several studies on the effects of robots on employment have pointed to a positive or neutral impact. For example, Graetz and Michaels (2018) observed a positive impact of robots on productivity in the EU, the US, and the UK, with neutral effects on employment. UNCTAD (2017) noted that the relationship between robot adoption and employment varies by country, but that countries that had a higher increase in the use of robots also experienced an increase in manufacturing employment. Klenert et al. (2023) did not find that robots negatively impacted the share of low-skill workers across Europe. However, other studies found negative effects. For instance, Giuntella et al. (2022) studied the impact in China, finding negative employment effects.

⁴ When considering trade, the literature has focused on the potential of AI to reshape trade. For example, Agrawal, Gans and Goldfarb (2018) highlight the role of AI as a general-purpose technology with the capacity to enable new forms of cross-border services, especially in sectors such as finance, healthcare, and digital services. Furman and Seamans (2019) emphasize the role of AI in enhancing the ability of firms to compete globally, which can lead to the reallocation of resources and changes in comparative advantage across countries. Baldwin (2020) emphasized the potential of AI to affect global value chains by reducing the need for low-cost offshore production.

due to higher profits in less competitive markets), neoclassical (reduction in wages due to displaced workers result in new equilibrium and full employment), Keynesian mechanism (increase in income due to workers appropriating gains of the increase in productivity resulting in increased wages, consumption and employment), and Schumpeterian mechanism (product innovation creating demand for labour). The authors note the challenge of considering all these compensatory mechanisms in assessing the impact of technological change on jobs: “Indeed, only an account of the full thread of sectors linked by technological and demand flows - explicitly accounting for the distribution of diverse innovative opportunities and demand elasticities - would do justice to the complexities of the compensation question.”

Vivarelli (2014) notes two types of analysis on the impact of technological change on jobs considering compensatory mechanisms: the study of the impact quantitatively on the total number of jobs, and the analysis of the impact of technological change on different types of workers, either by the level of skill or by the level of the routinization of the work. He examines the compensatory mechanisms and highlights that the introduction of new products (Schumpeterian mechanism) is the most powerful way to counterbalance labour-saving tendencies of process innovation. He also notes that the compensatory mechanisms are only partial and are dependent on historical processes and institutional aspects. In relation to the impact of technological change on developing countries, Vivarelli notes that the main channels of effect are trade and foreign direct investment and technology transfer, which depend on the absorptive capacity of countries.

Dosi et al. (2019) note that the results of the theoretical and empirical literature on the effects of technological change on jobs depend on 1) the different levels of aggregation in empirical and theoretical studies: firms, sectoral, macro-level, 2) the consideration of the effects of technological change that is embodied (e.g. investment in new machinery) and disembodied (via R&D expenditures), 3) the consideration of labour-saving effects and compensatory mechanisms, and 4) the relative balance between process and product innovation. I would add to this list the consideration of the effects of trade and changes in specialization patterns in open economies, which affects jobs in one country even when technological change is happening in other countries.

One macro-level model that considers many compensatory mechanisms is the model of structural economic dynamics by Pasinetti (1993). This labour-only multi-sectoral economic model considers that a household sector provides labour for the many production sectors in the economy (each producing one type of product) and receives income for that work, which is used to purchase the products produced for consumption. Products are produced in each production sector with different labour productivity, which change over time at different rates for each sector. The rates of consumption per capita for each product are also different and change at different rates. The model considers that it is possible to reach saturation of the demand for the different products, and it also assumes that new sectors could be created (product innovation). Therefore, the structure of the economy in terms of output and employment changes over time. The changes in productivity and demand, as well as the changes in the number of sectors, are considered exogenous in the model. Some of the conclusions of Pasinetti's model are: 1) the effect of technological change on jobs in a sector depends on the balance between the change in labour productivity and the change in demand caused by the reduction in the price of the product and the increase in income; 2) technological change affects different sectors at different rates and the aggregated effect depends on the sum of the changes in demand and productivity; 3) with the saturation of demand, in the long term, without the creation of new sectors, the tendency is for technological progress to reduce employment; 4) the creation of new sectors can serve to counterbalance the tendency to reduce employment. Therefore, increasing productivity when facing saturation of demand would lead to technological unemployment. The ways to counterbalance are the reduction of hours worked, but more importantly, the creation of

new sectors (products). Thus, economic diversification is the way to avoid technological unemployment.

As mentioned above, economic models in the Neoclassical tradition assume that the impact of technological change on jobs is the reduction of wages due to the displacement of workers, which results in a new equilibrium and full employment. Thus, these models assume that there would be, in principle, no long-term technological unemployment and focus on the impact of technological change on the change in the labour share in national income. Acemoglu and Restrepo (2018a) note that the neoclassical models that assume technological change as factor augmenting (increasing the productivity of either capital or labour) predict that productivity increases will increase value-added, which increases demand for labour.

Acemoglu and Restrepo (2019) propose a new task-based framework in which production requires the completion of a range of tasks, some of them performed by capital in terms of machines (automation) and others by labour.⁵ Technological change in one sector can affect the initial division between capital and labour in two ways. It can increase the set of tasks performed by machines (automation) or create new tasks for labour (increasing the labour content). Automation affects labour demand through the result of productivity effect (increases in the value-added of production and the demand for the labour related to non-automated tasks) and displacement effect (labour displaced out of the tasks that were automated). The authors argue that it is not the transformative technologies that produce high increases in productivity that harm labour, but the technological change that has low productive effects that do not counterbalance the displacement effects. The negative effect of “so-so technologies” is higher in the context of lower wages and abundant labour (e.g. low-income developing country), with a low increase of productivity by automation compared with higher displacement, than when automation occurs in the context of higher wages and scarce labour (e.g. high-income industrialized countries). The second way technological change affects labour is through the introduction of new tasks, which impact jobs via the productivity effect, as in the case of automation, and the reinstatement effect (the absorption of labour in the new tasks). The authors also consider that technological change can be factor augmenting without changing the task content in a sector. In this case, either labour or capital becomes more productive in their tasks, and the net effect on jobs is the result of the productivity effect and the substitution effect (substitution between labour- and capital-intensive tasks within a sector due to task prices). The authors then consider the multi-sector economy and include a composition effect on the impact of automation, new tasks and factor augmenting on jobs. This effect is the result of the absorption of labour displaced in one sector to other sectors due to changes in demand (input-output changes and demand elasticities of different products). When considering open economies, the composition effect also captures the effects of trade and changes in production and export patterns. The authors use this framework to examine the impact of technological change on the US economy since World War II. Acemoglu and Restrepo (2019) argue that their conceptual framework offers two lessons: 1) the fact that automation could reduce labour because they could bring higher displacement effects than productivity effects, 2) given the displacement effect, increases in wages may not be at the same rate as increases in productivity.

Interestingly, the task-based framework arrives at the same conclusion as Pasinetti's (1993). In Acemoglu and Restrepo's (2019) framework, automation replaces tasks performed by labour, and if the increase in productivity is not high enough to compensate for the displaced labour, the result is an increase in productivity with a reduction of employment and wages. The solution is the creation of new tasks,

⁵ Task-based frameworks are usually more narrowly focused on labour markets and how individual tasks are transformed by technology. In contrast, structural models usually have a broader scope, considering the impact of technology sectors and the economy at large.

which can be seen as the creation of new sectors (goods/services). Therefore, this framework arrives at a conclusion, as [Pasinetti \(1993\)](#), that the creation of new tasks (diversification) is the way to address technological unemployment.⁶

Applying their task-based framework to examine the potential impact of AI, [Acemoglu and Restrepo \(2020a\)](#) point to the dangers of AI being applied in the “wrong” type of innovation, those that focus on automation and not on the creation of new tasks. They note the potentially negative social effects on jobs and inequality and highlight many reasons that indicate that the market mechanism does not guarantee the application of AI in innovation that creates new tasks. These reasons include the externalities created by innovation (which are not addressed by the market), competing technological paradigms, reduced role of the government in the US in innovation, the focus of leading tech companies on automation, tax incentives for upgradation and taxation of labour. The authors call for directionality in the innovation using AI towards the creation of new tasks in areas such as education and healthcare. In a subsequent article, the authors report the negative effects of automation contributing to the increase in wage inequality in the United States between 1980 and 2016 ([Acemoglu and Restrepo, 2022](#)).

The framework presented provides insights into the possible channels and directions of the impact of technological change on jobs but cannot be used to examine the potential impact of that change on the global economy or different groups of countries. Theoretical models do not necessarily seek to study the effects of technological change on jobs in developing countries and do not usually consider the trade channel.

Another aspect that is not emphasized in [Acemoglu and Restrepo \(2019\)](#) framework, as well as in the other strands of the literature mentioned above, is that diversification is path-dependent. This path dependence has been underscored by the strands of literature on economic complexity and economic geography, which differentiates between related and unrelated varieties of innovations, and thus diversification. [Hausmann and Klinger \(2006\)](#) introduced the concept of “product space,” a network where products are linked based on the likelihood of a country exporting both, suggesting shared capabilities. They also developed a “density” metric within this space to assess the average similarity between a potential new product and a country’s existing exports, with higher density indicating proximity to many exports and implying shared capabilities ([Hausmann and Klinger, 2007](#)). Subsequent research, following [Hidalgo et al. \(2007\)](#), has used network analysis to show how product space can direct economic structural changes, leading to faster transitions when countries move into denser areas. Research in economic geography has highlighted the difference between related and unrelated varieties of innovation and the significance of industry relatedness and diversification through branching into related industries ([Neffke and Henning, 2008](#); [Boschma and Frenken, 2011](#); [Content and Frenken, 2016](#)). Studies utilizing product space proximity, following [Hidalgo et al. \(2007\)](#), indicate that regions develop new industries that complement existing ones ([Neffke, Henning, and Boschma, 2011](#)). Areas with denser product spaces often see quicker economic growth and are more likely to develop new industries aligned with regional capabilities ([Boschma, Minondo, and Navarro, 2012, 2013](#)). This body of work found that regions tend to diversify through related innovation closely linked to existing regional capabilities, and that unrelated innovation, although fewer, trigger extended sequences of incremental innovations ([Boschma, 2017](#)).

The model used in this paper ([Freire, 2017, 2019](#)) considers: 1) the effect of technological change on jobs in countries at different levels of

income, 2) trade as a channel of impact of the adoption of AI in one country and sector into other countries, 3) endogenous diversification, and 4) path dependence in the process of diversification.

2.1. How does the literature characterize AI as different from other technologies?

In the model and analysis presented in this paper, an important question is how AI differs from other technologies, so those differences could be incorporated into the model and computer simulation. Based on a review of selected academic papers, books and articles on AI ([Table 1](#)), this paper identifies the following list of common elements that distinguish AI from other technologies:

- Automation of cognitive tasks: AI’s ability to automate tasks that require cognitive functions sets it apart from technologies that automate only routine or manual tasks.
- Learning and adaptability: AI’s ability to learn from data and adapt over time, unlike static technologies.
- General-purpose technology: AI as a technology with broad applications across multiple sectors, driving innovation in ways previous technologies could not.
- Enhanced decision-making: AI’s role in improving decision-making through better data processing and predictions.

Automation of cognitive tasks and learning and adaptability are the most frequently acknowledged characteristics across the literature

Table 1
Elements that distinguish AI from other technologies.

	Automation of cognitive tasks	Learning and adaptability	General-purpose technology	Enhanced decision-making
Acemoglu and Restrepo (2020b)	X			
Autor (2015)	X	X		
Autor, Mindell and Reynolds (2020)	X	X		
Autor and Salomons (2018)	X			
Baldwin (2020)	X			
Bessen (2019)	X			
Brynjolfsson and McAfee (2014)	X	X	X	
Brynjolfsson and McAfee (2017)	X	X	X	X
Chui, Manyika, and Miremadi (2020)	X	X	X	
Cockburn, Henderson and Stern (2019)	X	X	X	X
Frey and Osborne (2017)	X			
Kaplan and Haenlein (2019)	X	X	X	
Mokyr, Vickers, and Ziebarth (2015)		X		
Russell (2020)	X	X		
Tegmark (2020)	X	X	X	
Varian (2021)	X			
West (2020)	X			
Zuboff (2020)	X	X		

Note: Publications are listed in alphabetical order of the names of the authors.

⁶ Although task-based frameworks usually consider diversification mainly in terms of labour and job diversification within the existing economic structure, focusing on how workers transition between different types of tasks within and across industries. In contrast, evolutionary structural models consider diversification as a transformation of the economic structure itself, involving the creation of entirely new sectors and the resulting changes in existing ones.

surveyed. These aspects make AI unique in the effect of disrupting labour, surpassing previous waves of technological change that primarily affected physical labour. However, ultimately, the effect of these characteristics in the work in which AI is applied is the increase in productivity, saving labour. Therefore, despite the difference in the characteristics of AI as compared with other technologies, the effect on labour is the same of previous automation technologies. Enhanced decision-making is less cited in the literature surveyed, but its effect is the similar: improve productivity and save labour. The general-purpose technology aspect of AI, surprisingly, is also less cited. However, in this case, its effect is not on productivity but on the possibilities for the combination of AI with other technologies, enlarging the set of potential work that could be affected through increasing productivity. Therefore, the difference between AI and other technologies in this regard is related to the scope and the scale of the impact, affecting more tasks and more jobs.

3. Model

This section presents a brief description of the model used in this paper, proposed by Freire (2017, 2019). The analysis conducted in the paper uses the model without any alteration. A detailed presentation of the model is beyond the scope of this paper. For information, the appendix contains the equations of the model. Freire's model formalizes Pasinetti's (1993) theoretical framework of structural change and economic dynamics of open economies and adds endogenous technological change, change in consumption patterns, and diversification of economies.

In the model, many countries produce a variety of products (goods and services)⁷ and trade with each other. Labour is the only factor of production. The units of the analysis of the model are the sectors that constitute an economy. Each country's economy comprises one household sector, many production sectors, and one research and development sector (R&D) (Fig. 1).

The household sector comprises the country's population, provides labour (L) to other sectors, and consumes products (domestic or imports) (Table 2). Consumption coefficients (c) give the consumption per capita of each commodity. Labour is uniform in quality so that each unit of labour is equivalent as a means of production and remunerated by a uniform wage rate (w). Labour is mobile between production and R&D, but it is not mobile between countries (no migration). In each period and each country, a proportion of the population is engaged in the production sectors; the rest is either working in R&D or unemployed.

Each production sector produces one single good, and all products are final goods.⁸ Quantity of commodity j produced domestically in country k is equal to the domestic and foreign demands over all countries (number of countries is R):

$$Q_{j,k} = \sum_{h=1}^R c_{j,h,k} N_h \quad (1)$$

Labour productivity levels in each production sector are given by labour coefficients (l), which reflects the amount of labour required to produce one unit of product. The model assumes monopolistic competition in domestic and international markets, and markups (MK) for prices of the products. Thus, the price of commodity j produced in country k and consumed in country h is given by the labour coefficient to produce commodity j in country k multiplied by the wage rate in country k and the markup price of commodity j produced in country k and consumed in country h :

$$p_{j,h,k} = l_{j,k} w_k MK_{j,h,k} \quad (2)$$

A specific set of labour-embodied technologies characterizes each production sector. Technology is defined as a means to fulfil a human purpose, following Arthur (2011), and it can be a method or process.

Each country has one R&D sector in which workers search for combinations of technologies that result in products that fulfil human needs. The wages of the workers come from the production sectors through the price markups of their products. Thus, the sum of markups in the production sectors (SP) limits the number of workers in the R&D sector.

$$SP_k(t) = \sum_{h=1}^R \sum_{j=1}^m (l_{j,k}(t)(MK_{j,h,k}(t) - 1)Q_{j,k}(t)) \quad (3)$$

Another constraint is the number of people not employed in the production sector and who can join the innovation efforts. The model assumes that production sectors have priority in engaging workers before R&D sectors. The availability of labour to participate in R&D (LA) is given by:

$$LA_k(t) = L_k(t) - E_k(t) \quad (4)$$

The share of labour engaged in R&D (ϵ_k) is then given by:

$$\epsilon_k(t) = \frac{\min(SP_k(t), LA_k(t))}{L_k(t)} \quad (5)$$

Countries that are more productive than others can generate higher markup profits and, therefore, can support a larger share of the population engaged in R&D. Conversely, when the productivity of the country is low, R&D effort is compromised. Also, countries that have the most labour engaged in the productive sectors would have fewer people available for R&D.

All products are tradable, and countries can trade freely without trade costs. All output produced is consumed. Domestic production can be consumed domestically, exported, or both. Domestic consumption is the sum of the domestic consumption of the commodity locally produced and of that imported. Total exports of a given country do not need to match the country's total imports. The balance of payments of countries is not necessarily balanced at each period.

The model divides time into periods. Within each period, the model determines which country specializes in which products based on the demand, prices of products, and the amount of labour available for production. At each period (short-run) and country, the following state variables are given: labour coefficient of each sector, coefficient of consumption per capita of each commodity, markup prices, and wage. Prices, quantities produced, markups and wages are endogenous.

Nominal wage rates endogenously reflect the average productivity of the economy. At each period, the wage rate in other country k is given by the wage rate of country 1 multiplied by the ratio of average labour coefficients in both countries weighted by the employment shares:

$$w_k = w_1 \frac{\left(\sum_{j=1}^m l_{j,1} E_{j,1} / E_1 \right)}{\left(\sum_{j=1}^m l_{j,k} E_{j,k} / E_k \right)} \quad (6)$$

⁷ The model considers both goods and services; thus, it considers both manufacturing and services sectors. But the model does not differentiate between services and goods. Moreover, given that it does not allow for the movement of people between countries, the model only consider cross border trade in services in which a consumer in country A receives services from another country remotely, through the Internet, or post, for example. This would include consultancy reports, engineering, legal or medical advice, etc. However, the model does not include trade in services such as in the case of tourism, where a person as consumer or producer must move to another country.

⁸ By following Pasinetti's (1993) model, Freire's (2017, 2019) model consider that each sector only produces one product (good or services). Therefore, the model cannot explicitly consider intra-industry diversification, in which a sector produces products of different qualities and prices. However, such scenario is still considered in this model as similar products of different quality and price are regarded as being produced by a different set of technologies and thus being the result of different sectors.

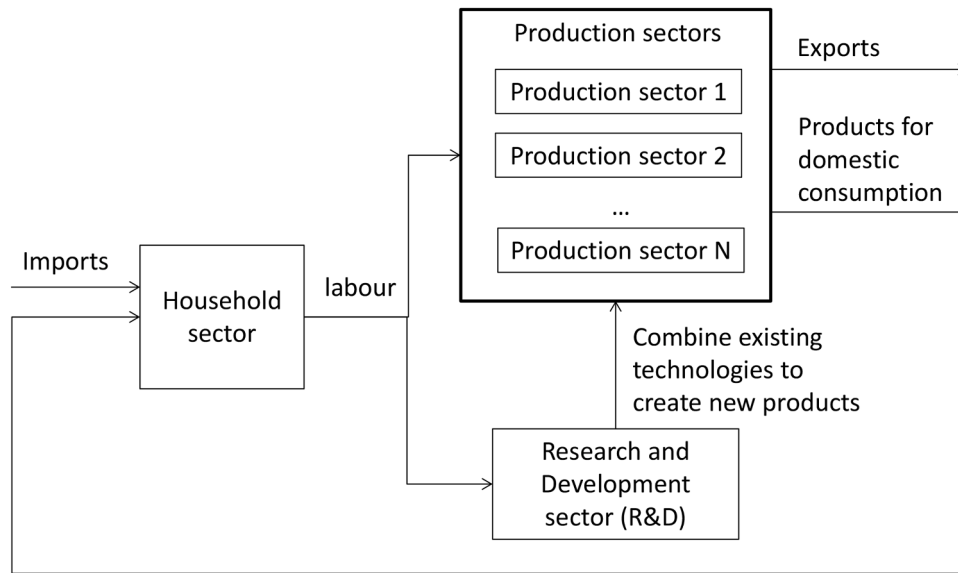


Fig. 1. Diagram of sectors within a country.
Source: Freire (2019).

Table 2
List of variables of the model.

Exogenous	
N_k	total population in country k
L_k	total labour available for production sectors in country k
State variables	
$l_{j,k}$	labour coefficient (labour input per unit of output) to produce commodity j in country k
$c_{j,k}$	coefficient of consumption per capita of commodity j in country k
$MK_{j,h,k}$	proportional markup added to the price of commodity j produced in country k and consumed in country h
w_k	nominal wage rate in country k
Endogenous	
$p_{j,h,k}$	price of commodity j produced in country k and consumed in country h
$c_{j,h,k}$	coefficient of consumption per capita in country h of commodity j produced in country k
$Q_{j,k}$	quantity of commodity j produced in country k
$E_{j,k}$	employment in sector j in country k
E_k	total employment in country k
$Y_{j,k}$	output of sector j in country k
Y_k	total output in country k
y_k	output per capita in country k
$Exp_{j,k}$	household expenditure in sector j in country k
Exp_k	total household expenditure in country k
exp_k	per-capita household expenditure in country k

The mechanism of setting markups is implemented in the model through a series of steps. First, initial markups are given for each market. The model considers that markup of exports are higher than for selling in the domestic market:

$$MK_{j,k,k}(0) < MK_{j,h,k}(0) \quad j = 1, \dots, m; h, k = 1, \dots, R; h \neq k \quad (7)$$

Sectors calculate their tentative prices (the labour component of the price may have decreased due to changes in the exchange rate, which affect the relative wage rates, or due to changes in the labour coefficient as a consequence of, for example, process innovation). Sectors of different countries compare prices and try to match the lowest tentative price by reducing their markups:

$$MK_{j,h,k}(t) = \min(p_{j,h,g}) / l_{j,k} w_k \quad j = 1, \dots, m; g, h, k = 1, \dots, R; g \neq k \quad (6a)$$

At the start of a period, sectors initially set their tentative markup for each market as the same as in the previous period:

$$MK_{j,h,k}(t) = MK_{j,h,k}(t-1) \quad j = 1, \dots, m; h, k = 1, \dots, R \quad (8)$$

Sectors for which the labour costs are higher than the lower price are forced out of the market.

In addition to the markup mechanism, which results in more than one country selling products for the same price, the model also accounts for the incomplete specialization of production and trade due to the limit in labour available for production. Therefore, the model does not assume ex-ante full specialization and allows for situations where similar products with different labour costs coexist in the global market.

In the long-term dynamics of the model, the economy changes with changes in consumption patterns and technical progress. However, different from Pasinetti's framework, both changes are endogenous to the model.

Similar to Pasinetti's framework, consumption patterns change according to a generalized version of Engel's law: (i) as incomes increase, there is a hierarchical order on the rate of satisfaction of needs, (ii) there are changes in consumption due to appearance of new products; (iii) there is a saturation of consumption. The saturation point of each product is not correlated with the hierarchy of essential goods. A less essential good could reach its saturation point earlier than an essential product. The appearance of a new product also changes the saturation point of the existing products. People may demand more and more of a product and less and less of others. The model also adopts the Keynesian view of consumer demand, in which households use a two-step decision process and decide on their demand for goods only after their actual incomes are known. First, households receive their income and, based on that and the current prices of products, decide on consumption preferences for the next period. If the income received is lower than the latest expenditure, then people choose to consume less. If, on the other hand, the income received is higher than the latest expenditure, then people will decide to consume more. When households consume in the following period, firms decide the level of employment to fulfil that demand, which determines income in the next period.

Regarding technological change, the model considers that all countries can perform product and process innovation and emulation.⁹ Process and product innovations, which create a product or process that is new to the world, are assumed to be less frequent events than emulation, which is an innovation that creates a product or a process that is new only to the country. Those who try to emulate have more information about the potential new product than those trying to create a new product for the world. Emulators may not know how to produce the new product initially, but they know the services it provides and the human needs it fulfils, and they know that there is a demand for the product.

In the model, the evolution of economies is path-dependent. The goods that a country can produce at any point affect what the country will produce next. In each period and each country, there is an adjacent possible of potential new sectors that could be created by the permutation of the existing set of technologies (Kauffman, 2010). Therefore, product innovation and emulation have to be part of the adjacent possible of the country. The model considers that some potential new products in the country's adjacent possible are not relevant solutions. They may be a combination of technologies, but they do not fulfil any human need. Therefore, only a subset of the adjacent possible would result in a new process or product through process innovation, product innovation, or emulation.

The R&D sector performs innovation and emulation in each country. The emergence of new processes or products in one country triggers the effort of emulation in other countries. There are also knowledge spillovers between sectors because the technologies used in the production of a product in one sector can be combined with technologies of different sectors to create new processes and products. The closer the economy is to full employment, the higher the effort towards process innovation to reduce the labour requirement in the existing production base. Similarly, the higher the level of unemployment, the higher the effort

towards product innovation to create new sources of demand and employment.

Of the number of people dedicated to process innovation, part of them works on process innovation new to the world. The remaining works on the emulation of process innovation that happened in other countries. The backwardness of the production base determines the allocation between these two groups. The higher the share of sectors that uses technologies that are not at the frontier (i.e., have lower productivity than the same sector in other countries), the higher the share of people dedicated to process innovation that emulates the most advanced technologies.

As for product innovation, the share of people dedicated to finding new to the world innovations and the people devoted to emulation is given by the economy's diversification. If the economy is less diversified, there are many opportunities for imitating the products that already exist in more diversified countries. Thus, more people will be engaged in the emulation process than in creating products that are new to the world. On the other hand, if an economy is already much diversified, there are fewer products in the world that are not already produced by that economy. As a consequence, fewer people will be engaged in the emulation of production, and more people will be trying to innovate. Thus, the higher the diversification, the higher the share of people involved in product innovation instead of emulation, and vice versa. The rate of product emulation is also affected by the demand for the product. Sectors that are rapidly expanding attract more emulation efforts than slow-growth or declining sectors.

3.1. Technology in the model

Technology is explicitly presented in the model proposed by Freire (2017, 2019) as labour-embodied technologies required to produce each product. In Freire's pure labour economy, labour-embodied technologies are defined following Arthur (2011) as a set of methods, heuristics, and procedures to produce a good or service.¹⁰ Each production sector i is characterized and can be represented by a set τ_i of n technologies ($\tau_i = \{t_1, t_2, t_3, \dots, t_n\}$) and two sectors i and j are set apart by the set of technologies used in the production:

$$\tau_i \neq \tau_j \quad i, j = 1, 2, \dots, m, \quad i \neq j \quad (7a)$$

For example, a particular technology t_x could be a procedure composed of operations (tasks) such as "move to location A," "grab the object," "lift the object," "turn to the direction of location B," and "drop the object." It could also be an algorithm to solve an equation or a heuristic to take a decision in the context of limited information.

3.2. How AI is considered in this paper

In this paper, I assume that AI can be a technology that is a part of the set of labour-embodied technologies required to produce a product. Clearly, AI is not a labour-embodied technology; it requires a computer to run. However, I assume that AI can be seen as a technology that has the capacity to perform cognitive tasks of labour-embodied technologies as well or better than any person in the model. Therefore, AI is considered a technology that emulates or surpasses a person in its capacity to use and act on labour-embodied technologies (e.g. methods, heuristics, procedures, decisions, etc). Whenever AI is added to the set of technologies of a sector, it can perform some of the other labour-embodied technologies that characterize that sector, effectively augmenting the productivity of workers of that sector. I also assume that AI is more efficient than humans in performing those other labour-embodied technologies, therefore, it reduces the labour coefficients of

⁹ Given that the model does not differentiate between goods and services, it does not consider any difference between process and product innovation between sectors that produce goods or services.

¹⁰ In a model with capita and labour, technologies would also include tangible tools such as gadgets and devices such as computers, industrial robots and other machines.

the production of sectors in which it is introduced.

For example, suppose a sector i is characterized by the set of technologies $\tau_i(0) = \{t_1, t_2, t_3, \dots, t_n\}$ in time 0, and the technology AI (t_{ai}) is added to that set in time 1, $\tau_i(1) = \{t_1, t_2, t_3, \dots, t_n, t_{ai}\}$. I assume that the primary effect of that change is the reduction of the labour coefficient of sector i ($l_i(1) < l_i(0)$) because AI will execute some of the other technologies $\{t_1, t_2, t_3, \dots, t_n\}$ instead of the workers in that sector.

Therefore, AI is assumed to be different from other labour-embodied technologies in the model because instead of being another technology in the sense of a method, heuristic, algorithm, procedure, etc, that is executed by a person in the model, AI acts as a person executing those other technologies.

For simplicity in the implementation of the model, I assume that AI is made available exogenously to all economies at the same time. Workers in the R&D sectors of each country will then be able to use AI in combination with the existing labour-embodied technologies available in the economy to find process or product innovations.

3.3. How AI in the model implement the characteristics of AI listed in section 2.1?

In the case of the characteristics of automation of cognitive tasks and enhanced decision-making, the AI in the model and simulations implements them through its effect in significantly increasing productivity, having the effect of reducing labour requirements by almost 100% in the sectors in which it is applied (simulations in Section 4.1). In this paper, I also consider the effect of AI in R&D by considering that AI increases the productivity of process and product innovation, augmenting labour in R&D by automating cognitive tasks in the R&D sector (simulations in Section 4.2).

The characteristic of general-purpose-technology is implemented in Section 4.3, with the increase of opportunities for combinations of technologies.

However, in the model and simulations, the characteristic of learning and adaptability is not replicated. The effect of AI on productivity is considered to happen instantaneously, not over time.

4. Simulation and results

To study the effects of AI in countries with different levels of development, I run computer simulations of the model with nine countries for a period of 200-time units.¹¹ All countries have the same characteristics at the beginning of the run. They start producing and trading the same seven products, have the same population size (100 people), and the same productivity, income and consumption levels. The stochastic nature of the innovation process is the element that creates different outcomes at the end of the runs. Fig. 2 illustrates a run of the computer simulation. Each line in the graph represents the evolution of GDP (in nominal terms) of a country in the simulation.

In each run of the simulation, I identify at period 100 (the middle of the simulation) the three countries with lowest GDP per capita (low-income countries), the three with the highest (high-income countries), and the three remaining countries (middle-income countries). I then add a technology representing AI that has the effect of reducing the labour required for production in the sectors that it is used. I run one benchmark scenario, without AI, and ten scenarios in which AI has the effect of reducing the labour required by 10%, 20%, 30%, 40%, 50%, 60%, 70%,

¹¹ I could have run the simulation with at least three countries, one each representing a low-, middle- and high-income country. I have used nine countries to have more variance in the results related to each one of these categories of countries, with three countries with lower incomes, three with middle incomes, and three with high incomes at the end of each round of simulation. A higher number of countries would increase the time required to run the simulation, which was already significant when using nine countries.

80%, 90% and 99%, respectively.

The set of initial configuration values and parameters for the benchmark scenario using the model proposed by Freire (2017, 2019) is listed in the appendix. I run 100 simulations of each of these scenarios (1100 in total). I then compare the average results for low, middle and high-income countries in terms of levels of employment, GDP, wage rates and diversification at the end of the simulation period. I also study the results of the model assuming the possible effects of AI in increasing the pace of product and process innovation through the complementarity effect between AI and cognitive labour in the R&D sector.¹²

4.1. Impact of AI in reducing labour requirement

The results of the simulations show a limited impact of the introduction of AI on aggregated GDP and employment (Fig. 3). The effect is positive in high-income countries and slightly negative in middle and low-income countries. In the aggregate, the impact is marginal when considering what is usually expected given the concerns with technological unemployment due to the adoption of AI, even when considering the more extreme scenarios of 90% and 99% of reduction of labour requirement in sectors using AI. Nevertheless, the different effects of AI on the GDP in low, middle and high-income countries, although small compared with the benchmark scenario, still result in increases in inequality between countries (Fig. 4).

Structural economic dynamics entails a constant shift of employment between production sectors. That is also the case with the introduction of AI, but the simulations also show that there are substantial shifts in the distribution of jobs between employment in the production sectors and the R&D sector (Fig. 5). In all countries, the first scenario (AI reducing the labour requirement by 10%) results in more jobs in the production and R&D sectors. In this case, technological unemployment in sectors that adopt AI is counterbalanced by the lower prices of products (owing to competition), which increases real income and the resulting demand, creating jobs. Moreover, higher profits in sectors that use AI increase investments in R&D and add jobs in that sector.

The other scenarios, those in which AI has a higher reduction effect on labour requirements, result in a reduction in jobs in production sectors. The capacity of countries to keep employment in the aggregate at about the same level would depend on their shift of jobs from the production to the R&D sector. In low-income countries, employment in the production and R&D sectors is lower for scenarios in which AI reduces labour requirements from 10% to 80%. For scenarios with higher impact on labour requirements, employment in production sectors further reduces while employment on R&D sectors becomes higher than in the benchmark scenario. Similar dynamics happen in middle-income countries, but in that case, employment is higher in productive and R&D sectors for scenarios up to 30%; employment reduces in both types of sectors in scenarios of 40%, to 60%, and employment is higher in R&D sectors for scenarios that have a reduction of more than 70% in labour requirement with the adoption of AI. As for the high-income country, employment in production and R&D sectors is higher for the scenario of 10% reduction and higher only for the R&D sector for the other scenarios. In these scenarios, the increase in employment in R&D in high-income countries compensates the jobs losses in the productive sectors.

The dynamics described above is the result of the interplay between the levels of profits in the sectors with AI and its result in investment in R&D, and the actual outcome of that investment in terms of new sectors created or technological progress in existing sectors, which depend on the productive technologies available in the countries and the competition between them in terms of prices of similar products. There seems

¹² Given that the paper applies the model by Freire (2019), robustness checks would test changes in parameters of that model. Luckily, Freire (2019) reports robustness checks changing the rate of product innovation and emulation and shows that it does not affect the results of the model.

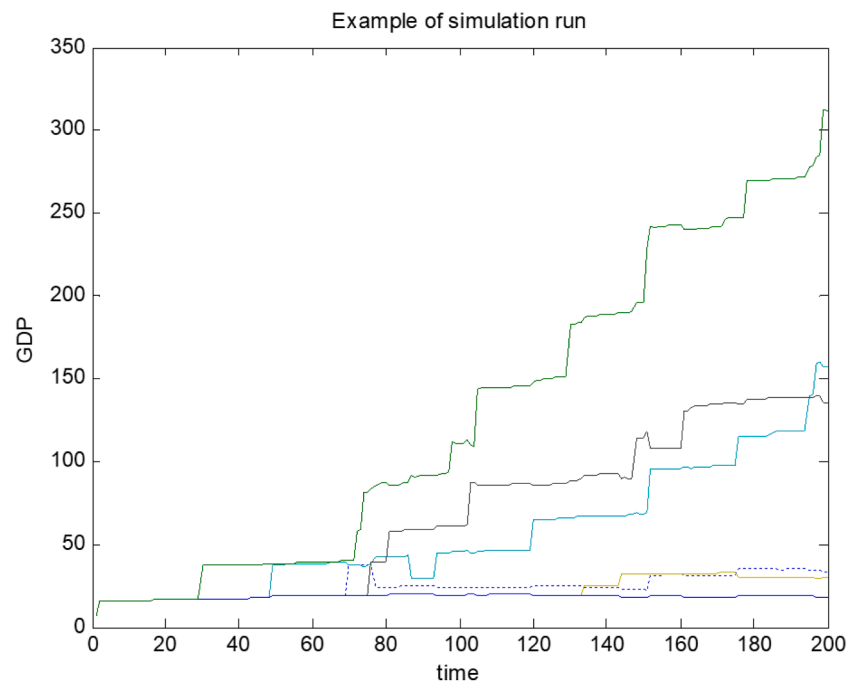


Fig. 2. Example of a simulation.

Note: Each line represents the evolution of GDP of one of the nine countries in the simulation. The figure only shows six lines because the bottom four countries have the same GDP at the end of this particular simulation.

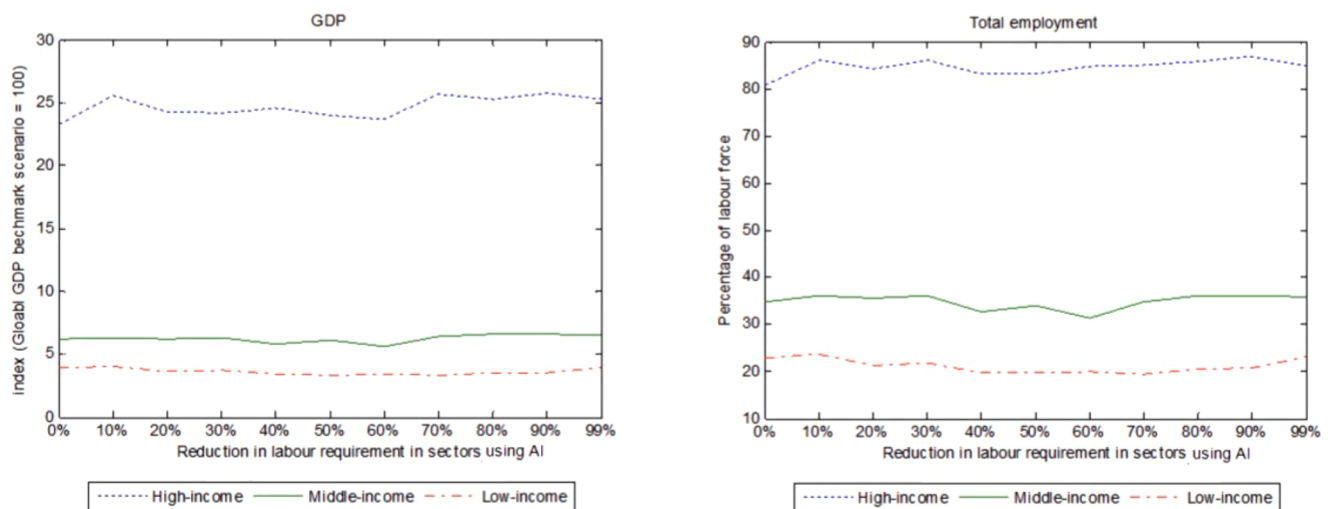


Fig. 3. Impact of introduction of AI on aggregated GDP and employment.

to be a level of investment in R&D for it to become effective, and in low-income countries that level is not reached in scenarios in which the effect of AI is to reduce labour requirement from 10% to 80%. In these scenarios, there are extra profits created in sectors that adopt AI. Still, they are not enough to generate the needed investment in R&D that creates new sectors and technological innovations at a pace faster enough to compensate for the losses of jobs in production sectors. That reduces the profits compared with the benchmark scenario and thus reduces also the employment in R&D. For scenarios in which the reduction in labour requirement is higher, the levels of profits are high enough for R&D to be effective, which generates more profits and more employment in the R&D sector. The range of ineffective investment in R&D is smaller for middle-income countries and does not exist for the average of high-income countries in the simulation experiments.

The results discussed in this section consider only the effect of AI in

reducing labour requirements for production. The next sections will consider other two potential effects: the increase in the pace of innovation due to complementarity between AI and R&D labour, and the increase of possibilities of combinations between AI and other technologies.

4.2. Impact considering a faster pace of innovation

The introduction of AI could be considered complementary to certain types of cognitive jobs. If that is the case with jobs in R&D, AI could increase the pace of innovation. For example, by enabling a more effective and efficient innovation process (Füller et al., 2022), identifying emerging trends and technologies (Mühlroth and Grottko, 2022), or enabling new methods of problem-solving and creating opportunities for innovation across a wide range of applications (e.g., Cockburn et al.,

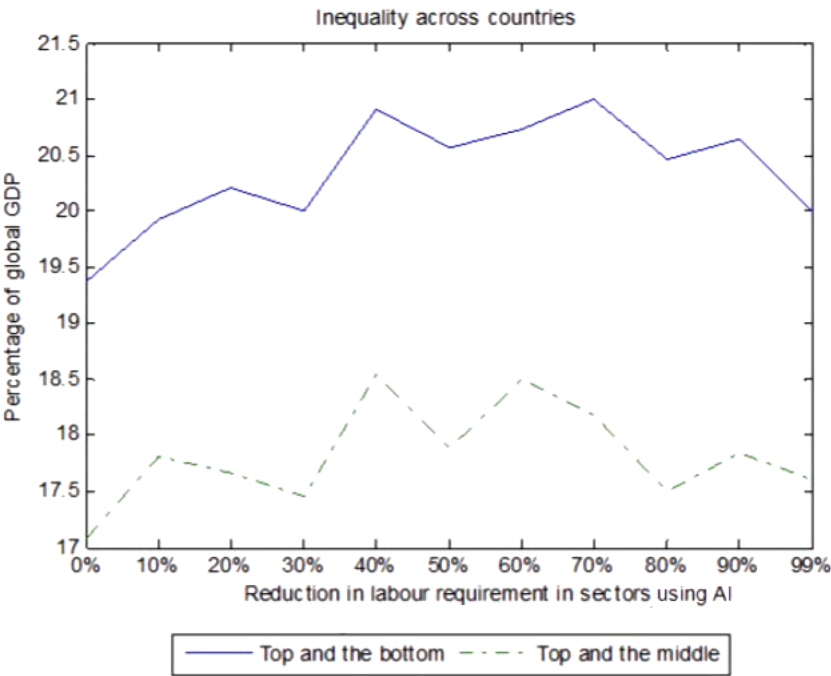


Fig. 4. Inequality between countries rises.

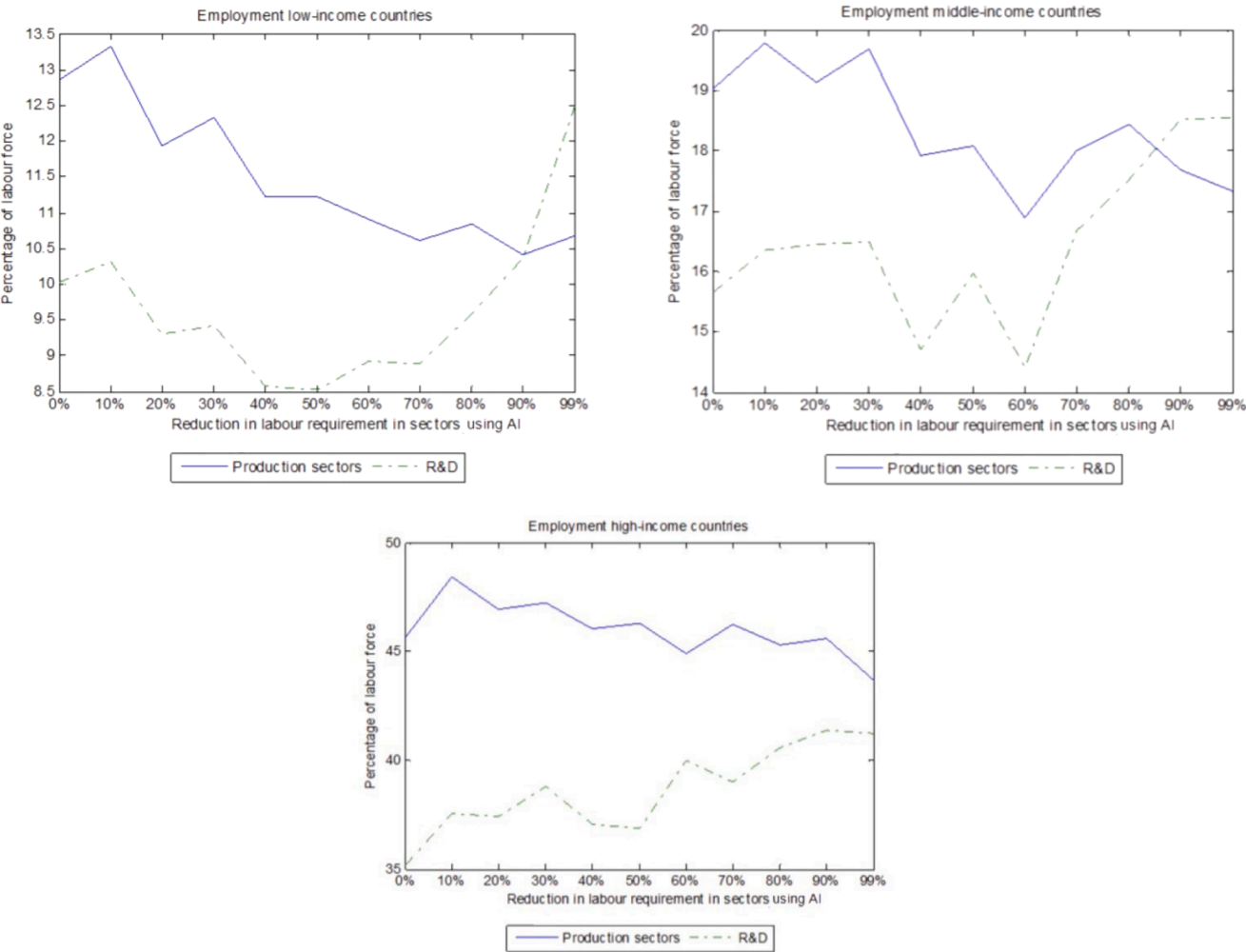


Fig. 5. AI causes a shift in the distribution of jobs.

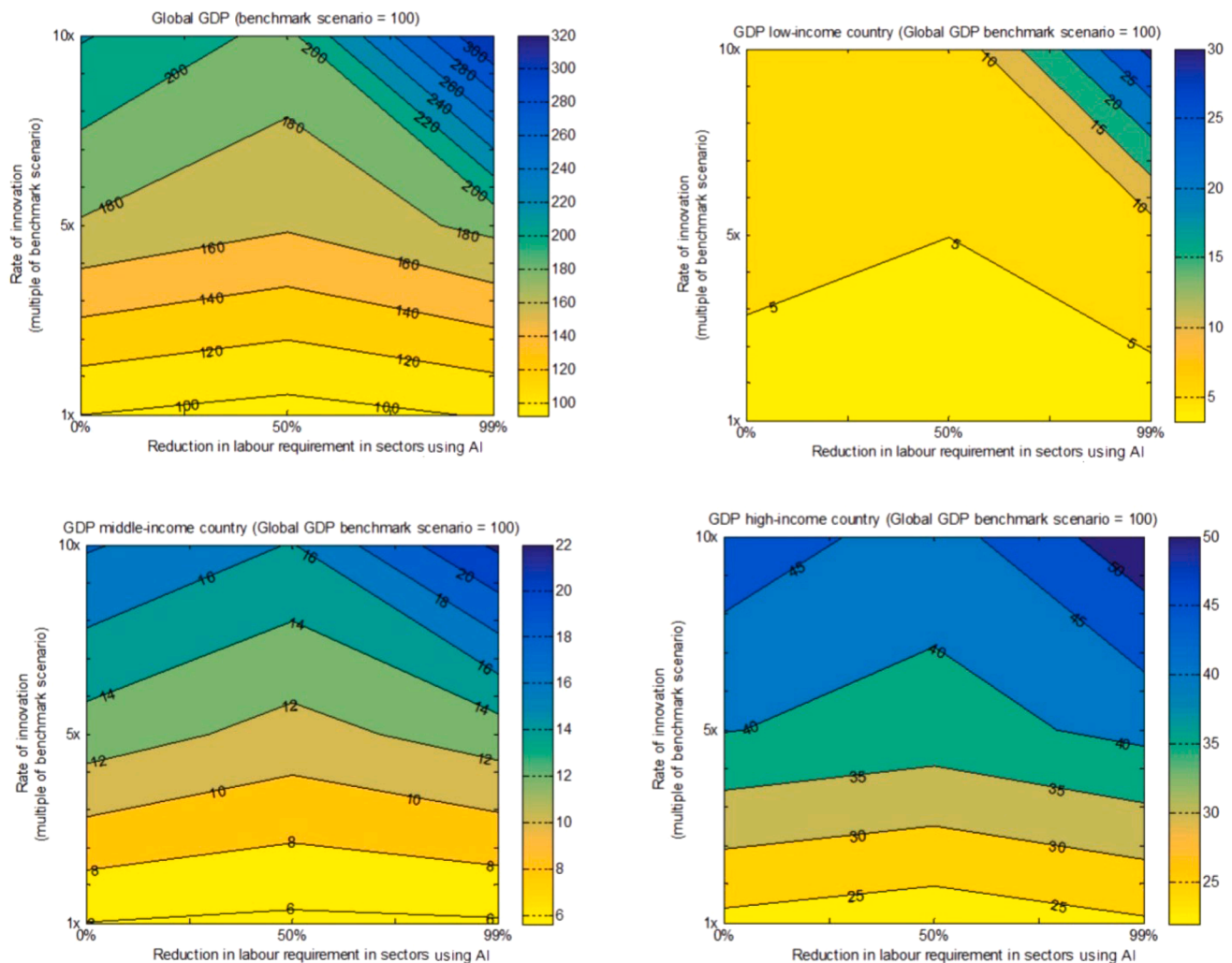


Fig. 6. Reduction of jobs in sectors adopting AI and faster innovation: Impact on aggregated GDP.

2019). This section discusses the results of computer simulations of the model that consider the effects of that increase in innovation combined with the reduction of labour required with the adoption of AI.

I use scenarios in which different levels of reduction of labour requirements are combined with innovation that happens five or ten times faster than in the benchmark scenario. I assume that the pace of both product and process innovation could increase with the adoption of AI. These have different immediate effects on jobs in production sectors. Process innovation reduces the labour requirements of an existing sector, which result in workers being laid off of their jobs. In the case of product innovation, a new sector emerges, which create jobs; even if a few, as in the case of when AI is adopted and reduces labour requirements by 99%.

The average of the results of the 100 runs for scenarios considered in the analysis is presented in the figures in this section. They show the pace of innovation in the vertical axis, which increases from the bottom (benchmark) to the top (ten times faster than in the benchmark). On the horizontal axis, the figure shows the reduction of labour required with the adoption of AI. It increases from the left (0%) to the right (99%). The contour lines in the graph join points that represent combinations of these two dimensions that result in the same values of the economic variable considered. Different colours in the figure represent different economic outcomes.

Fig. 6 shows the results concerning the impact on aggregated GDP. The first graph shows that there is a significant increase in global GDP with the rise in the pace of innovation. That is expected, considering that

technological innovation is the key driver of economic development. However, the increase is not linear when considering the impact of AI in reducing the labour required. When AI reduces the need for labour by 50%, it results in lower global GDP when compared with the benchmark scenario, while reductions close to 99% result in higher global GDP. A similar pattern is seen in relation to the GDP in low, middle and high-income countries. The reason for that pattern is the same discussed in the previous section. For intermediate values in the reduction of labour requirements, the increased profits in sectors that adopt AI and the resulting investment in R&D are not enough to create new sectors at the pace required to compensate for the low employment and lower prices of products, when compared with the benchmark scenario.

One interesting result of this analysis is that it shows low-income countries catching up and even surpassing middle-income countries in terms of total GDP when assuming the high pace of innovation and high reduction of labour requirements in sectors that adopt AI. This is the result of technological leapfrogging. In this scenario, low-income countries engage a sizeable share of the labour force in R&D and innovate mainly by combining AI with other technologies. Middle-income countries, on the other hand, have a higher set of technologies related to the previous technological paradigms and a sizeable share of their innovation does not use AI. That difference enables low-income countries to catch-up.

These dynamics are reflected in the resulting inequality across countries, as shown in Fig. 7. Inequality between high-income and low and middle-income countries increases with a faster pace of innovation

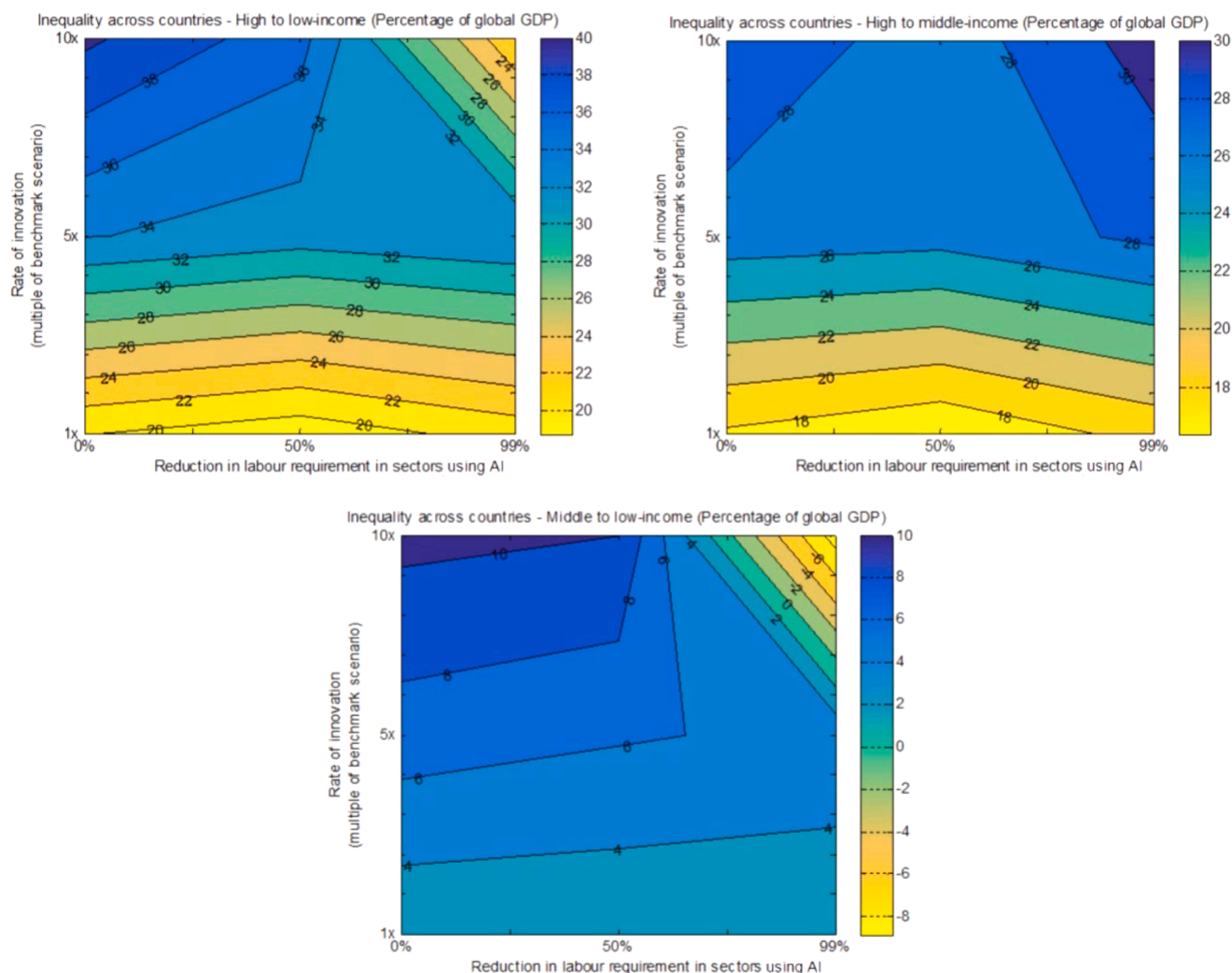


Fig. 7. Reduction of jobs in sectors adopting AI and faster innovation: Inequality across countries.

because the former can benefit the most, given its technological leadership and a higher share of the labour force engaged in R&D. This increases the chances of a successful product and process innovation in these countries. However, for even faster paces of innovation, low-income countries can leapfrog and innovate mostly using AI, which reduces their distance to high and middle-income countries.

The impact on total employment is shown in the graphs in Fig. 8. They show that employment increases with the increase in the pace of innovation. Economic diversification that results from the emergence of new sectors creates new jobs in the aggregate. However, as in the case of total GDP, the effect on employment is not linear in relation to the reduction of labour required in sectors that use AI. When the reduction is 50%, the level of total employment is lower than in the case of the benchmark scenario, while it is higher for the instance of reductions of 99% of the labour required. Although low-income countries leapfrog and catch up with middle-income countries for faster paces of innovation combined with a higher reduction of labour required in using AI, in terms of total employment, substantially fewer jobs are created when compared with high and middle-income countries. This indicates a situation of jobless growth in which the increase in GDP outpaces the rise in the number of jobs in low-income economies.

Fig. 9 shows the shifts in the distribution of jobs from production sectors to R&D. The number of jobs in production sectors increases in high, middle and low-income countries with the increase in the pace of innovation. This is the result of economic diversification due to product

innovation. However, the increase is smaller the higher the impact of AI in reducing the need for labour in the sector in which it is adopted. As for the number of jobs in the R&D sector, in low and middle-income countries they also increase with the increase in the pace of innovation but at a significantly slower pace for scenarios in which AI reduces labour requirement by up to 50% when compared with the scenarios that reduce labour requirement by closer to 99%. For high-income countries, the number of R&D jobs is reduced with the pace of innovation because of the limited size of the labour force, given the number of people already engaged in the production sectors. In a model with labour mobility across countries (i.e. migration), the number of jobs in the production sector and R&D could continue to increase at the same time.

In summary, the analysis of the model suggests that if AI complements the cognitive labour of R&D workers, increasing the pace of product and process innovation, the effects would be mainly positive in terms of aggregated global GDP, and GDP for low, middle and high-income countries. It could even allow for the catching up of low-income countries due to leapfrogging. Total employment would also increase in comparison with the benchmark scenario. However, the inequality across countries in most scenarios would still increase, and the leapfrogging of low-income countries would be accompanied by a more modest increase in job creation. The shift in the distribution of jobs between production and R&D sectors would continue as in the results of the analysis of scenarios considering the effect of AI only on the reduction of labour requirement.

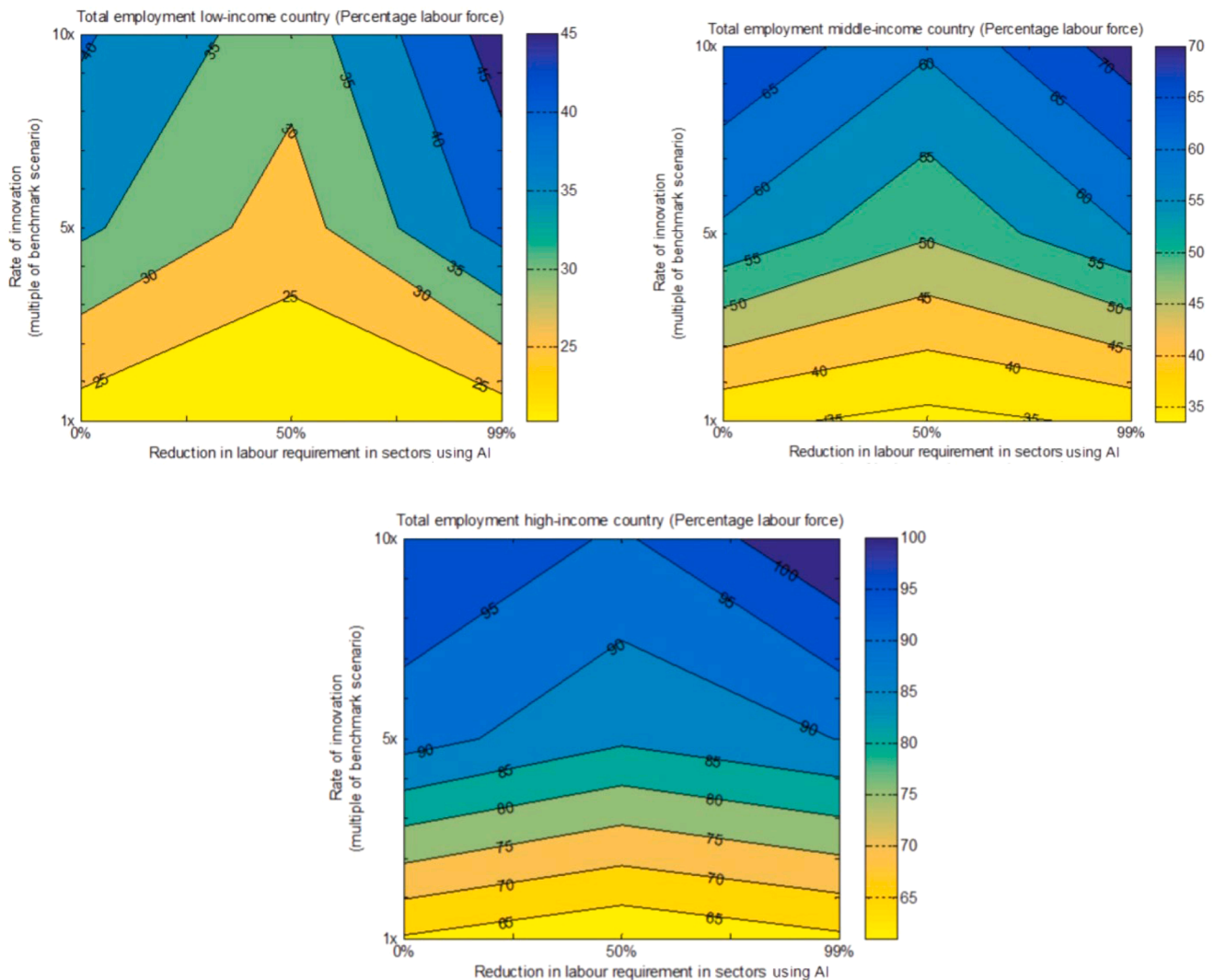


Fig. 8. Reduction of jobs in sectors adopting AI and faster innovation: Impact on employment.

4.3. Impact considering more opportunities for the combination of technologies

This section discusses the results of computer simulations considering the possibility that, in addition to the effect of AI in reducing labour requirements, it also increases the opportunities of combination with other technologies, which would increase the pace of innovation. I consider the benchmark scenario in which an innovation that is a combination of AI with other technologies has 20% chance of being useful and in becoming a new product or process. I also consider two other scenarios in which that probability is 40% and 60%, respectively.

Fig. 10 shows the results of the computer simulations in terms of GDP. The graphs show that the effect is smaller than in the case discussed in the previous section related to the complementarity between AI and R&D labour.

In fact, the impact of the reduction of labour requirement seems to dominate the results, as opposed to the effect of the combination of technologies that could facilitate innovation. The latter effect is also not linear. Increasing the probability of successful combinations increases the level of GDP up to a point, after which it reduces. The reason is that more successful combinations imply that a larger number of traditional sectors undergo process innovation that reduces the number of jobs. When that is not accompanied by enough product innovation that creates jobs, the result is lower GDP and employment. This result is more

pronounced in high-income countries, those that are more successful in innovating.

The impact of this dynamics in the inequality across countries is that it mostly reduces along the innovation dimension when comparing high-income to middle and low-income countries (Fig. 11). The result is more complex when considering inequality between middle and low-income countries and would depend more strongly on the effect of AI in reducing labour requirements. The reason is that in these scenarios the middle-income countries are not in disadvantage in terms of having a larger set of products under the previous technological paradigm. Given that the probability of successful innovation using AI is higher than those that do not include AI, most of the innovations in these countries will have AI as in the case of the low-income countries. In this case, the larger set of technologies of middle-income countries, as well as the higher number of people engaged in R&D, is an advantage to these countries.

The impact of more combinations of technologies on total employment is shown in Fig. 12, and the shift of jobs between production and R&D sectors is presented in the graphs in Fig. 13. Usually, more combinations are associated with more jobs than in the benchmark scenario. However, the result depends on the number of AI sectors created and the reduction of labour that AI adoption results. The interplay between these elements shows a complex pattern, but the impact in the aggregate is not high.

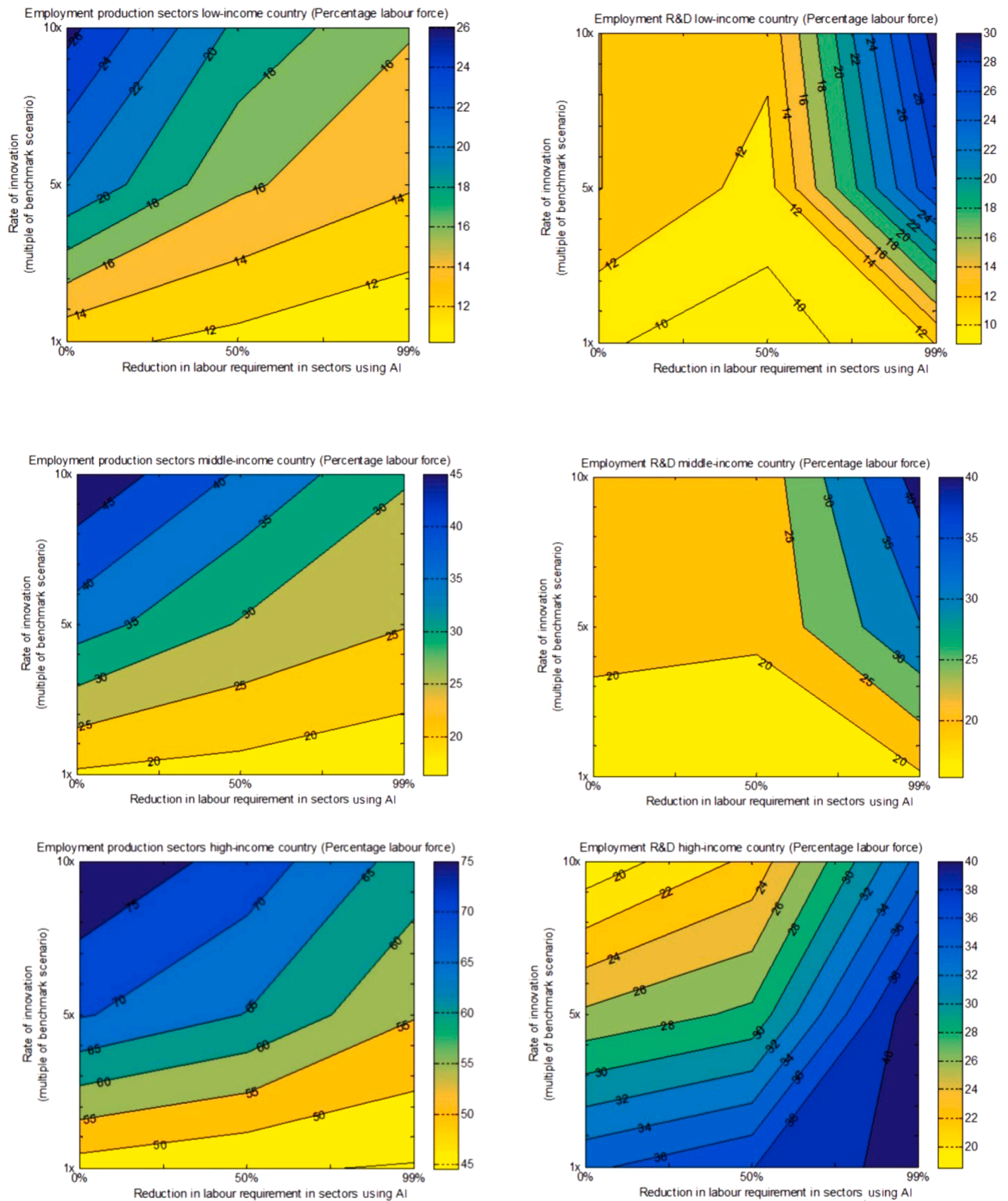


Fig. 9. Reduction of jobs in sectors adopting AI and faster innovation: Shift on the distribution of jobs.

5. Discussion

The results of the computer simulations of the model show that the economic system could be very robust to AI in relation to the impact on the aggregated GDP and employment. This result is aligned with studies of this strand of the literature that suggest that the total number of jobs is

unlikely to be affected by AI in the aggregate, but AI will change the tasks performed by people and contribute to inequality (e.g., Autor, 2015, and Arntz et al., 2017). The result regarding the effect of AI on total employment is in line with the findings of Vermeulen et al. (2018), which also adopts an evolutionary and multi-sectoral economic framework to assess the potential impact of AI and robots on employment and

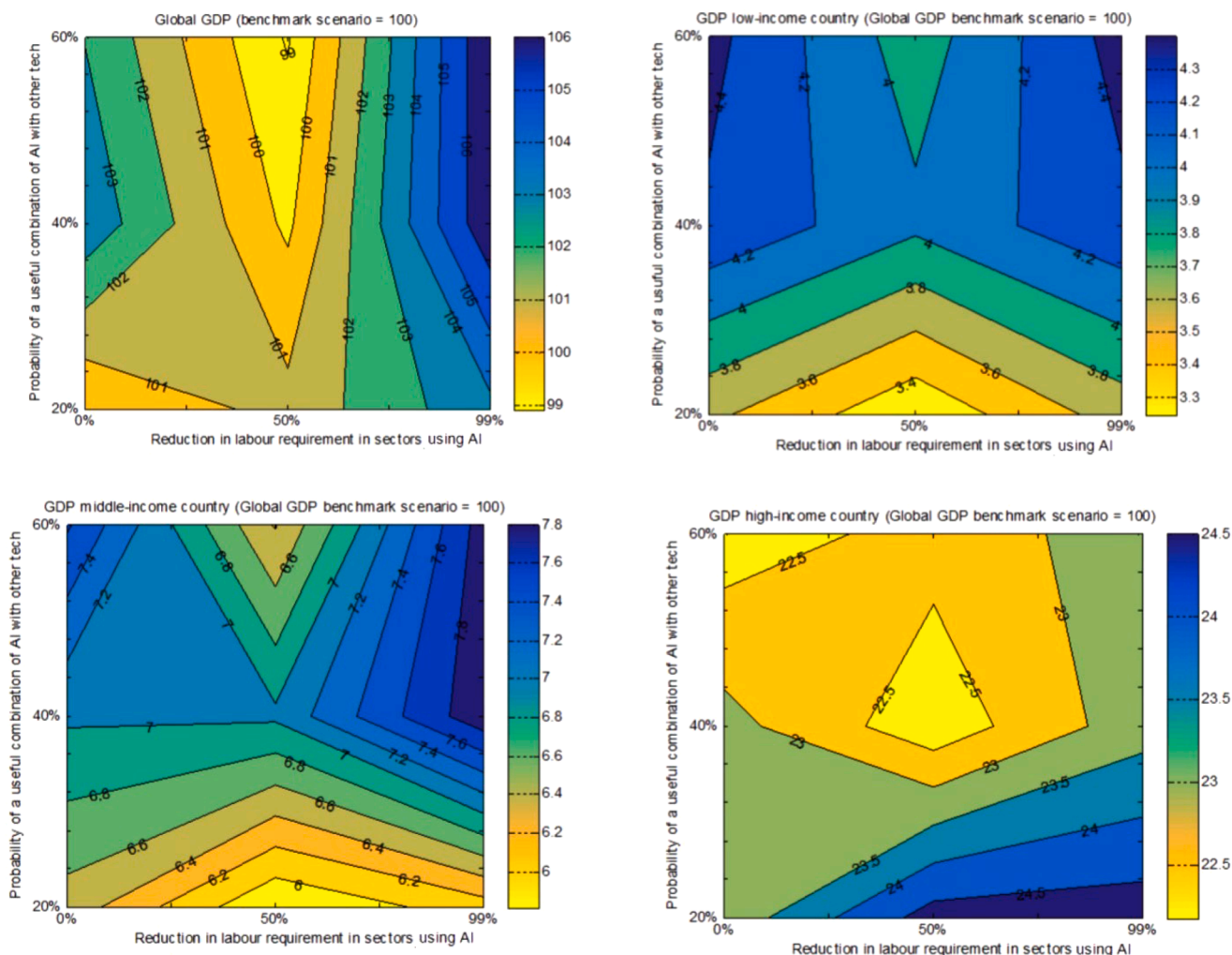


Fig. 10. Reduction of jobs in sectors adopting AI and more opportunities for combination with other technologies: Impact on aggregated GDP.

concludes that mankind is facing the usual effects of structural change instead of mass technological unemployment.

The results presented also suggest that despite the small change in the aggregate, in general, high-income countries benefit and middle and low-income countries are affected negatively, even if only slightly. As a result, across country inequality still increases in most scenarios. This suggests that the effect of AI may not be different from the impact of previous technological revolutions and that catching up by middle and low-income countries would continue to require the active and strategic action of governments and the private sector in facilitating the increase of productive capacities of economies.

Nevertheless, the analysis also shows that the potential of AI to reduce labour requirements to practically zero could produce results that are indeed very different from reductions of lower magnitude. There seems to exist a non-linear effect of reducing labour requirement that is related to the interplay between the levels of profits of sectors that adopt AI, their use to invest in R&D, and the effective outcome of the innovation efforts, which depend on the technological set available in the country for combination into new products and processes and the number of people engaged in R&D. These factors, combined with the changes in prices due to changes in wages rates and labour productivity, and the competition between countries in the global markets create complex outcomes that show the difficulty to assess the impact of technological change without considering the structural economic dynamics.

The discussion in the literature concerning the impact of AI on

different types of workers has concentrated on skills and levels of routinization. This paper, similar to [Dosi et al. \(2019\)](#) and [Acemoglu and Restrepo \(2019\)](#) highlights the impact and the importance to consider the interrelation between the innovators and production workers, but also notes that the effects are different for countries at different levels of income.

When considering the effects of relaxing key assumptions of the model, I find that the potential negative effects on low-income countries could be accentuated. For example, the simulations assume that all countries have access to AI simultaneously in the middle of the simulation run and that all countries can immediately adopt and adapt the technology in new products and processes. However, countries gain access to technology at different times and the capacity to adopt, adapt and disseminate a technology depend on the country's national system of innovation, which in the case of AI would include, among others, the existing capacity of firms in AI and the existence and access to big data relevant to the area of application and to be used in the machine learning. Low-income countries are at a disadvantage in that regard.

Another assumption is the mobility of labour between production sectors and R&D sectors. In reality, that mobility is hardly the case, particularly in poorer countries. In addition to institutions that facilitate labour mobility, this would require a labour force prepared in terms of skills to make this shift. Those employed in the R&D sectors would be searching for potential new products and for technological improvements in existing productions in a broader and expanding technological space, which would require even more skills, perhaps across many

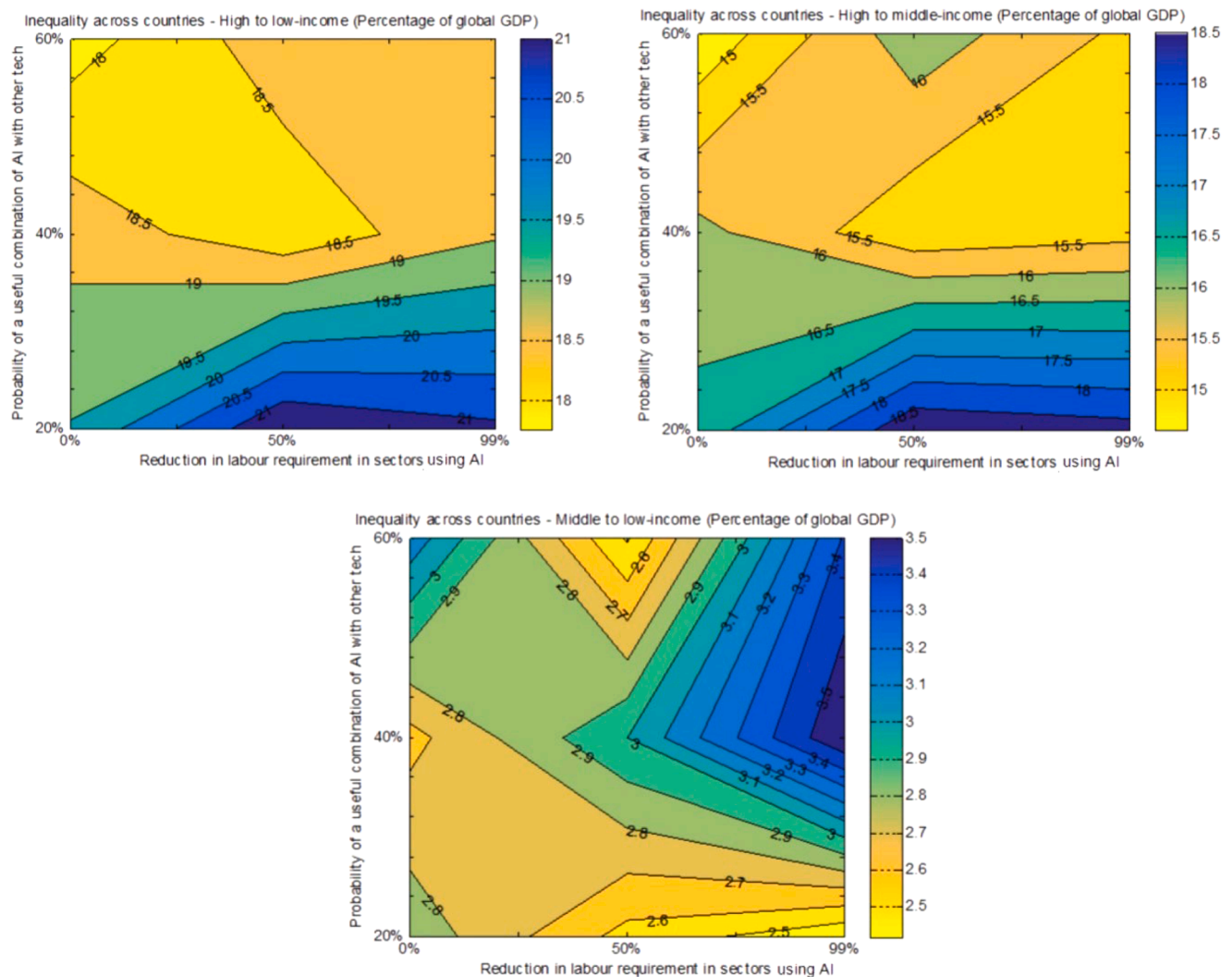


Fig. 11. Reduction of jobs in sectors adopting AI and more opportunities for combination with other technologies: Inequality across countries.

domains. All that shows the need for an educated and skilled labour force that is able and has the opportunity for continuous skills improvement.

The model also assumes that all profits in the production sectors are invested in R&D efforts. This assumption has a clear effect on the results given that the counteracting effect of product innovation depends on the number of people engaged in R&D. Although the concept of R&D in the model is broader than the usual and could easily be better approximated by the efforts of the national innovation systems, the assumption of the high priority of investment in innovation is unrealistic, particularly in poorer countries. In low-income countries, the domestic industry is characterized by a large share of micro, small and medium-sized enterprises (SMEs) with low capacity of investment, and those industries that are established through foreign direct investment (FDI) usually invest in process innovation and revert to profits to their countries of origin. Moreover, [Dosi et al. \(2019\)](#) notes that the investment channel is undermined by financialization (i.e. profits are invested in speculative products the financial markets instead of on new productive investments or R&D). The model used in this paper relies on the investment channel in which profits are used to fund R&D. Therefore, financialization can impact that mechanism, reducing the counterbalance mechanism of creation of employment in R&D.

Another area of future research is to include considerations regarding the different effects of process and product innovation in manufacturing and services sector in the economies in the model. For

example, the results of the current analysis show a shift of jobs from production to R&D sectors and if we consider a services sector with lower labour productivity and lower opportunities of process and product innovation the results may provide some information about job polarization in low, middle and high-income countries, with the disappearance of jobs in the production sectors and increase in services and R&D.

6. Conclusions

This paper studied the impact of AI in low, middle and high-income countries considering structural economic dynamics, trade and endogenous technological change that includes product innovation and emulation. The results show that this emergence of this new technological paradigm would only have a marginal impact on the aggregated macroeconomic outcomes in terms of GDP and employment, but would result in increasing inequality across countries. There would be, furthermore, a significant shift in the distribution of jobs between production and R&D sectors. The impact would be negative for countries that are not able to adopt, disseminate and adapt AI to create new production sectors as well as those that are not able to facilitate the movement of workers between production and R&D sectors.

This paper contributes to the literature on the impact of AI on employment and inequalities. The findings regarding AI's minimal impact on total employment are consistent with the literature on

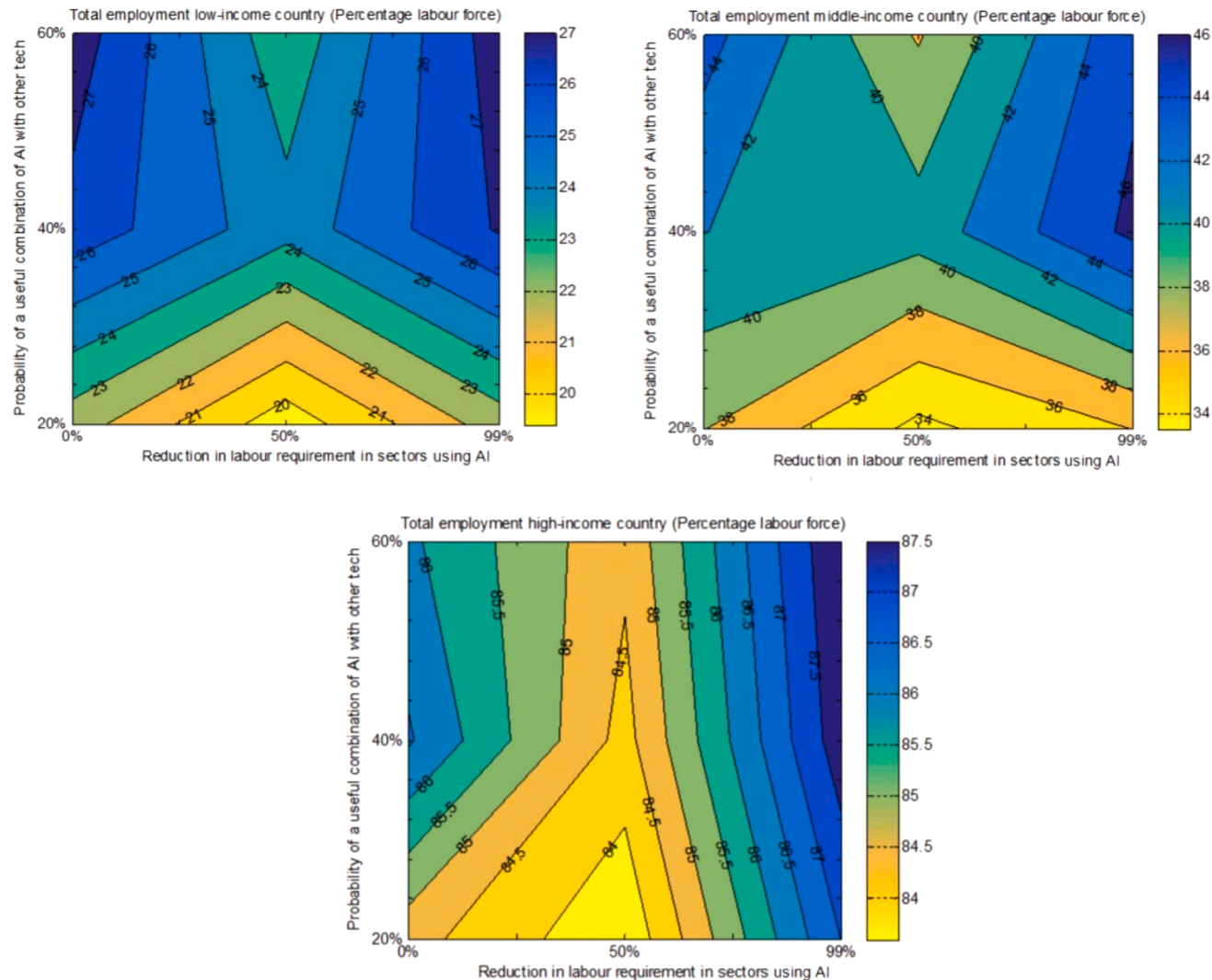


Fig. 12. Reduction of jobs in sectors adopting AI and more opportunities for combination with other technologies: Impact on employment.

technological change and labour markets (e.g., Autor, 2015), which highlights that while technological advancements, including AI, may displace jobs in certain sectors, they often result in new job creation in other areas. This aligns with the model's outcomes, where AI-induced productivity gains lead to shifts in sectoral employment but do not significantly alter aggregate employment levels. However, the results on inequality suggest a more nuanced effect, as AI appears to widen the income gap both within and between countries.

Future analysis using this model could consider the representation of AI's adaptability and the timing of its adoption across countries, which are not fully captured in the current simulations. Future analysis could also consider the effect of trade facilitation in increasing the pace of product and process emulation, which are main strategies of poorer countries to catch up with countries at the technological frontier. Another area of future research is to include the services sector in the economies of the model. The results of the current analysis show a shift of jobs from production to R&D sectors, and if we consider a services sector with lower labour productivity, the results may provide some information about job polarisation in low, middle and high-income countries, with the disappearance of jobs in the production sectors and increase in services and R&D.

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The author has no relevant financial or non-financial interests to disclose.

Declaration of generative Ai and AI-assisted technologies in the writing process

During the preparation of this work the author used Grammarly to improve language and readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

Author statement

I confirm that the submitted work is original, has not been published elsewhere, and is not under consideration by another journal.

I declare no relevant financial or non-financial interests to disclose.

I declare that during the preparation of this work the author used Grammarly to improve language and readability. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

I confirm that the computer code used in this analysis and resulting

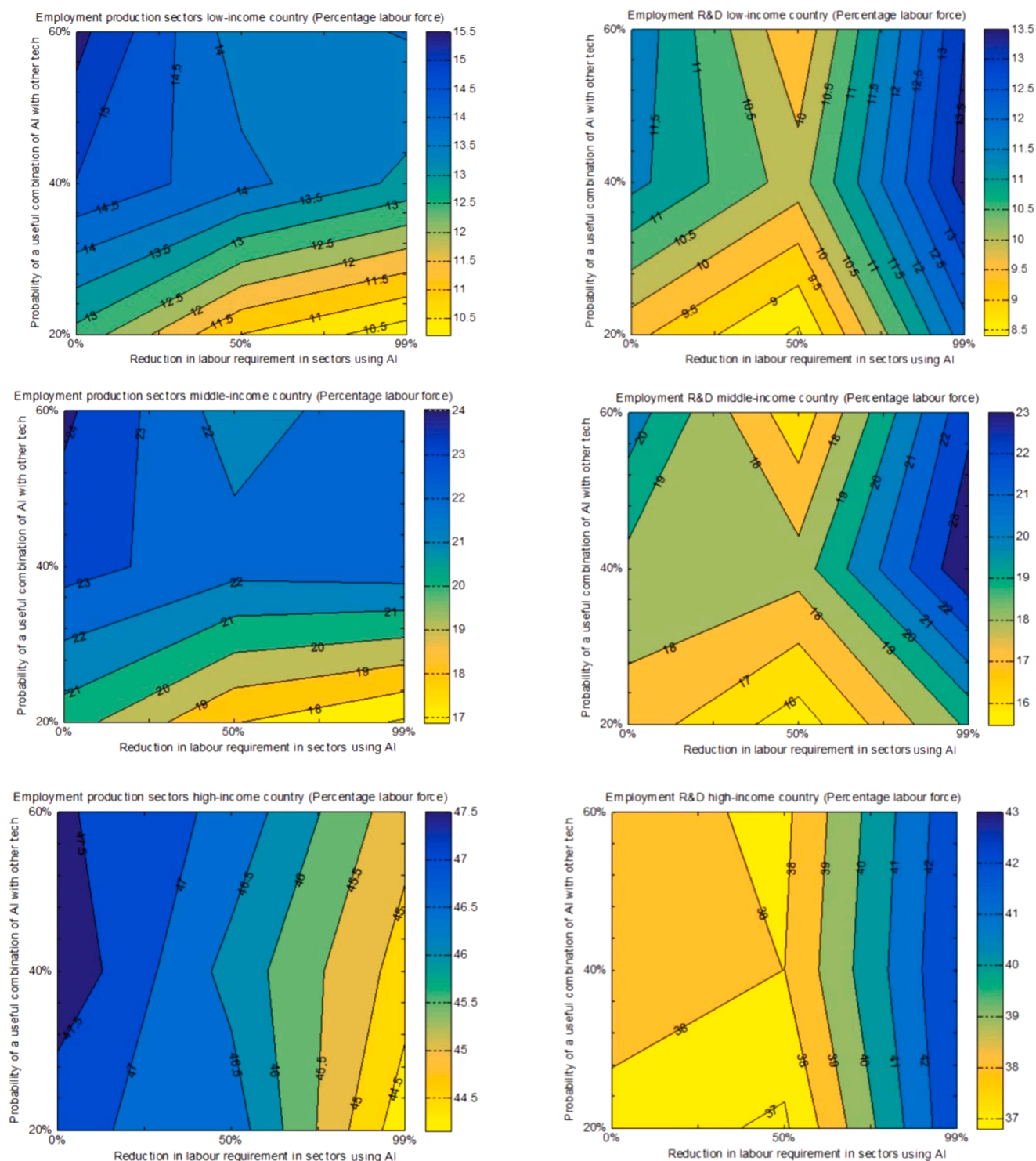


Fig. 13. Reduction of jobs in sectors adopting AI and more opportunities for combination with other technologies: Shift on the distribution of jobs.

data is available upon request to support the transparency and reproducibility of their work.

CRedit authorship contribution statement

Clovis Freire: Writing – review & editing, Writing – original draft,

Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Appendix

The model (Freire, 2017, 2019)

Short-run: Economic variables calculated based on exogenous and state variables

Prices: Prices reflect labour costs and markups

$$p_{j,h,k} = l_{j,k} w_k MK_{j,h,k}$$

Quantities:

Quantity of commodity j produced domestically in country k is equal to the domestic and foreign demands

$$Q_{j,k} = \sum_{h=1}^R c_{j,h,k} N_h$$

Calculate domestic and foreign demand using the following linear programming

Minimizes expenditure given consumer preferences (under the following constraints)

$$\text{Minimize } \sum_{k=1}^R \left(\sum_{j=1}^m \left(\sum_{h=1}^R c_{j,h,k} N_h p_{j,h,k} \right) \right)$$

$$c_{j,h,k} \geq 0$$

$$c_{j,k} = \sum_{h=1}^R c_{j,h,k}$$

Constraint: The consumption per capita of a commodity j in country k is the sum of the domestic consumption of commodity j that is locally produced and of commodity j that is imported

$$\sum_{h=1}^R \left(\sum_{k=1}^R c_{j,h,k} N_h \right) = \sum_{k=1}^R c_{j,k} N_k$$

Constraint: The sum of quantity produced of a commodity j in all countries is the same as the sum of the quantity consumed of that commodity in all countries

$$\sum_{j=1}^m l_{j,k} \left(\sum_{h=1}^R c_{j,h,k} N_h \right) \leq L_k$$

Constraint: The maximum total labour employed in each country is lower or equal than the total labour available in that country

Endogenous macroeconomic variables

For each country k , employment in each sector j is equal to the labour required to produce the quantity of commodity j produced domestically. Total employment is the sum of employment in each sector of the economy

$$E_{j,k} = l_{j,k} Q_{j,k}$$

$$E_k = \sum_{j=1}^m E_{j,k}$$

The output by sector ($Y_{j,k}$) is given by the price of the commodity multiplied by the quantity produced

$$Y_{j,k} = \sum_{h=1}^R c_{j,h,k} N_h p_{j,h,k}$$

Total output of the economy k is the sum of the outputs of the individual sectors

$$Y_k = \sum_{j=1}^m Y_{j,k}$$

Dynamics

Nominal wage rates endogenously reflect the average productivity of the economy. At each period, the wage rates in other countries are given by the wage rate of country 1 multiplied by the ratio of average labour coefficients in both countries weighted by the employment shares.

$$w_k = w_1 \left(\frac{\sum_{j=1}^m l_{j,1} E_{j,1} / E_1}{\sum_{j=1}^m l_{j,k} E_{j,k} / E_k} \right)$$

Change in markup prices:

The mechanism of setting markups is implemented in the model through the following algorithm:

1. Initial markups are given for each market. We consider that markup of exports are higher than for selling in the domestic market:
 $MK_{j,h,k}(0) < MK_{j,h,k}(0) \forall j=1, \dots, m; h,k=1, \dots, R; h \neq k$
2. At the start of a period, sectors initially set their tentative markup for each market as the same as in the previous period:
 $MK_{j,h,k}(t) = MK_{j,h,k}(t-1) \forall j=1, \dots, m; h,k=1, \dots, R$
3. Sectors calculate their tentative prices (the labour component of the price may have decreased due to changes in the exchange rate, which affect the relative wage rates, or due to changes in the labour coefficient as a consequence of, for example, process innovation).
4. Sectors of different countries compare prices and try to match the lowest tentative price by reducing their markups:
 $MK_{j,h,k}(t) = \min (p_{j,h,g}) / l_{j,k} w_k \forall j=1, \dots, m; g,h,k=1, \dots, R; g \neq k$
5. Sectors for which the labour costs are higher than the lower price are forced out of the market.

Technological change

Adjacent possible:

The possible new products and processes that can be introduced through innovation have to be part of the set of possible combinations of existing technologies in the economy (the adjacent possible). The algorithm used to determine the adjacent possible for each country k at the start of each period is the following:

Calculation of share of labour dedicated to innovation

Share of labour engaged in R&D

$$\epsilon_k(t) = \frac{\min (SP_k(t), LA_k(t))}{L_k(t)}$$

$$LA_k(t) = L_k(t) - E_k(t)$$

$$SP_k(t) = \sum_{h=1}^R \sum_{j=1}^m (l_{j,k}(t) (MK_{j,h,k}(t) - 1) Q_{j,k}(t))$$

$$\partial_k^{\text{product}}(t) + \partial_k^{\text{process}}(t) = 1$$

Out of the group of people engaged in R&D, a share of them is devoted to research towards a new product ($\partial_k^{\text{product}}$) (either to the country or to the world) and another share is devoted to finding a new and more productive way to produce an existing product ($\partial_k^{\text{process}}$)

(continued on next page)

(continued)

Shares of research dedicated to finding a new product or a new process are endogenous to the model and a function of the share of the labour force that is employed

Out of the group of people engaged in finding a new product, a share of them is devoted to research towards product innovation ($\sigma_k^{product}$) and another share is devoted to emulation ($\sigma_k^{product_emulation}$)

Shares of research dedicated to product innovation and emulation are endogenous to the model

Out of the researchers working to discover a more productive process of production, a share is devoted to research towards process innovation ($\sigma_k^{process}$) and another share is devoted to process emulation ($\sigma_k^{process_emulation}$)

Shares of research dedicated to process innovation and process emulation are assumed to be endogenous

Product and process innovation and emulation

Process innovation: In each country k the outcome of the work of one person engaged towards process innovation takes the form of a Poisson process with the arrival rate of the new process given by $\lambda_k^{process}$

Process emulation: One person engaged towards process emulation in country k would find a new process at an arrival rate of $\lambda_k^{process_emulation}$ that takes the form of a Poisson process. A sector will only undergo process emulation if the sector is lagging behind the technological frontier. When process emulation happens in a sector j , we consider that the sector adopts the technologies of the frontier country

Product innovation: In each country k the outcome of the work of one person engaged towards product innovation takes the form of a Poisson process with the arrival rate of the new sector given by $\lambda_k^{product}$. Labour coefficient of the new sector (l_{new}) is given by the average of the labour coefficients of the production sectors in activity in the economy in the previous period

Product emulation: In each country k the outcome of the work of one person engaged towards emulation takes the form of a Poisson process with the arrival rate of the new sector given by $\lambda_k^{emulation}$. For the process of emulation, it is necessary that a new product has emerged in another country. Labour coefficient at the time of the emulation (t') is the same as the labour coefficient of that sector at the time that it was created (t') in the economy of country h that initially introduced the commodity

Change in consumption patterns

Commodities' order from those that satisfy the most to those that satisfy the least essential needs: We enforce a decreasing order on the rate of consumption changes ($r_{j,k}$) in each country in each period
Saturation of demand: For each commodity j there is a maximum amount for the consumption per capita of that commodity given by $maxc_j$

Dual-decision hypothesis:

Households receive their income and, based on the current prices of products, decide on consumer preferences for the next period. If the income received is lower than the latest expenditure, people will have a lower expectation related to the extent to which their incomes will be able to fulfil their consumption in the next period, and they would decide to consume less. If, the income received is higher than the latest expenditure, people would decide to consume more. When households actually consume in the following period, firms decide on the level of employment to fulfil that demand, which determines income in the next period.

Variation in the composition of consumption may occur as a consequence of the introduction of new products. s_{ij} is the substitution and complementarity effect of the emergence of the new product i on the consumption of an existing commodity j

In each country k , by the time of the introduction of the new product i , the coefficient of consumption of existing product j is affected in the following one-time change

$$\partial_k^{process}(t) = E_k(t)/L_k(t)$$

$$\partial_k^{product}(t) = 1 - E_k(t)/L_k(t)$$

$$\sigma_k^{product}(t) + \sigma_k^{product_emulation}(t) = 1$$

$$0 \leq \sigma_k^{product}(t) \leq 1$$

$$0 \leq \sigma_k^{product_emulation}(t) \leq 1$$

$$\sigma_k^{product}(t) = \frac{m_k(t)}{m(t)}$$

$$\sigma_k^{product_emulation}(t) = 1 - \frac{m_k(t)}{m(t)}$$

m_k is the number of types of commodities produced in country k and m is the combined number of different types of commodities traded by all countries

$$\sigma_k^{process}(t) + \sigma_k^{process_emulation}(t) = 1$$

$$0 \leq \sigma_k^{process}(t) \leq 1$$

$$0 \leq \sigma_k^{process_emulation}(t) \leq 1$$

$$\sigma_k^{process}(t) = TF_k(t)/m_k(t)$$

$$\sigma_k^{process_emulation}(t) = 1 - TF_k(t)/m_k(t)$$

TF_k is the number of sectors in production in country k that are operating at the technological frontier, meaning that they have the highest productivity when compared with similar sectors in other countries

$$X_{j,k} \sim P(\epsilon_k \partial_k^{process} \sigma_k^{process} N_k(E_{j,k}/E_k) \lambda_k^{process})$$

$$l_{j,k}(t) = rdc_{j,k} l_{j,k}(t-1)$$

$$0 < rdc_{j,k} < 1$$

$$rdc_{j,k} \sim U(\beta_1, 1)$$

$$0 < \beta_1 < 1$$

Where β_1 is a parameter of the model

$$X_{j,k} \sim P(\epsilon_k \partial_k^{process} \sigma_k^{process_emulation} N_k(E_{j,k}/E_k) \lambda_k^{process_emulation})$$

$$l_{j,k}(t) = l_{j,h}(t)$$

$$X_k \sim P(\epsilon_k \partial_k^{product} \sigma_k^{product} N_k \lambda_k^{product})$$

$$l_{new,k}(t) = \frac{\sum_{j=1}^{m_k} l_{j,k}(t-1)}{m_k(t-1)}$$

$$\varphi_{j,c}(t) = \frac{\sum_{k=1}^R Y_{j,k}(t-1) - \sum_{k=1}^R Y_{j,k}(t-2)}{\sum_{i=1}^m \sum_{k \neq c}^R Y_{i,k}(t-1) - \sum_{i=1}^m \sum_{k \neq c}^R Y_{i,k}(t-2)}$$

$$X_k \sim P(\epsilon_k \partial_k^{product} \sigma_k^{emulation} N_k \varphi_{j,c} \lambda_k^{emulation})$$

$$l_{j,k}(t') = l_{j,h}(t')$$

$$r_{1,k} \geq r_{2,k} \geq \dots \geq r_{m_k,k}$$

$$\forall j \exists \max c_j, c_{j,k}(t) = \min(c_{j,k}(t-1)e^{\theta_{j,k}}, \max c_j)$$

$$\max c_j > MAX_j$$

$$MAX_j = \max(c_{j,1}(1), c_{j,2}(1), \dots, c_{j,m_k}(1))$$

$$\max c_j \sim U(MAX_j, \beta MAX_j)$$

$$\begin{cases} r_{j,k}(t) > 0, & \text{if } \exp_k(t-1) < y_k(t-1) \\ r_{j,k}(t) = 0, & \text{if } \exp_k(t-1) \geq y_k(t-1) \end{cases}$$

$$\begin{cases} r_{j,k} \sim U(0, \max(r)), & \text{if } \exp_k(t-1) < y_k(t-1) \\ r_{j,k} \sim U(-\max(r), 0), & \text{if } \exp_k(t-1) \geq y_k(t-1) \end{cases}$$

$$r_{j,k} = 0, \text{ if } \exp_k(t-1) = y_k(t-1)$$

$$c_{j,k}(t') \begin{cases} > 0, & \text{if } y_k(t'-1) \geq \alpha \\ = 0, & \text{otherwise} \end{cases}$$

$$s_{ij} \sim U(1 - \kappa, 1 + \kappa)$$

$$c_{j,k}(t) = \min(s_{ij} c_{j,k}(t-1)e^{\theta_{j,k}}, \max c_j)$$

$$\max c_j(t) = s_{ij} \max c_j(t-1)$$

Values and parameters for the benchmark scenario

The following is the set of initial configuration values and parameters for the benchmark scenario using the model proposed by (Freire, 2017, 2019):

- labour coefficient $l_1 = l_2 = \dots = l_9 = \{1, 1, \dots, 1\}$;
- coefficient of consumption per capita $c_1 = c_2 = \dots = c_9 = \{0.01, 0.01, \dots, 0.01\}$;
- wage rate in country 1 $w_1 = \$ 1$;
- $\beta = 2$; saturation of consumption not higher than 100% above the initial consumption per capita;
- $\lambda_k^{product} = \lambda_k^{process} = \lambda_k^{product-emulation} = \lambda_k^{process-emulation} = 1/500$; in any country k ($k=1,2,3,\dots,9$) each researcher finds a new product or a new process on average once every 500 periods;
- $u = 0.2$; 2 out of 10 potential new products invented as a combination of existing technologies are useful.

Data availability

The computer code used in the computer simulation can be made available upon request.

References

- Acemoglu, D., Restrepo, P., 2017. Working Paper No. 23285.
- Acemoglu, D., Restrepo, P., 2018a. Modeling automation. In: AEA Papers and Proceedings, pp. 10848–10853.
- Acemoglu, D., Restrepo, P., 2018b. The race between man and machine: implications of technology for growth, factor shares, and employment. *Am. Econ. Rev.* 108 (6), 1488–1542.
- Acemoglu, D., Restrepo, P., 2019. Automation and new tasks: how technology displaces and reinstates labor. *J. Econ. Perspect.* 33 (2), 3–30.
- Acemoglu, D., Restrepo, P., 2020a. The wrong kind of AI? artificial intelligence and the future of labour demand. *Cambridge J. Regions, Econ. Soc.* 13 (1), 25–35.
- Acemoglu, D., Restrepo, P., 2020b. Robots and jobs: evidence from US labor markets. *J. Polit. Econ.* 128 (6), 2188–2244.
- Acemoglu, D., Restrepo, P., 2022. Tasks, automation, and the rise in U.S. wage inequality. *Econometrica* 90, 1973–2016.
- Agrawal, A., Gans, J.S., Goldfarb, A., 2018. *The Economics of Artificial Intelligence: an Agenda*. University of Chicago Press.
- Arntz, M., Gregory, T., Zierahn, U., 2017. Revisiting the risk of automation. *Econ. Lett.* 159157–159160.
- Arthur, W.B., 2011. *The Nature of Technology: What it is and How it Evolves*. Free Press, New York.
- Autor, D.H., 2015. Why are there still so many jobs? the history and future of workplace automation. *J. Econ. Perspect.* 29 (3), 3–30.
- Autor, D.H., Mindell, D., Reynolds, E.B., 2020. *The Work of the Future: Building Better Jobs in an Age of Intelligent Machines*. MIT Work of the Future Report.
- Autor, D.H., Salomons, A., 2018. Is automation labor-displacing? productivity growth, employment, and the labor share. *Brookings Pap. Econ. Act.* 2018 (1), 1–63.
- Baldwin, R., 2020. *The Globotics Upheaval: Globalization, Robotics, and the Future of Work*. Oxford University Press.
- Barbieri, L., Mussida, C., Piva, M., Vivarelli, M., 2019. Testing the employment impact of automation, robots and ai: a survey and some methodological issues. DISCE - Quaderni del Dipartimento di Politica Economica dipe0006. Università Cattolica del Sacro Cuore, Dipartimenti e Istituti di Scienze Economiche (DISCE).
- Bessen, J.E., 2019. AI and jobs: The role of Demand. National Bureau of Economic Research (NBER) Working Paper No. 24235.
- Brynjolfsson, E., McAfee, A., 2014. *The Second Machine Age: Work, Progress, and Prosperity in a Time of Brilliant Technologies*. W.W. Norton and Company.
- Brynjolfsson, E., McAfee, A., 2017. The business of artificial intelligence: What it can—and cannot—do for your organization. *Harv. Bus. Rev.* Available at: <https://hbr.org/2017/07/the-business-of-artificial-intelligence>.
- Boschma, R., 2017. Relatedness as driver of regional diversification: a research agenda. *Reg. Stud.* 51 (3), 351–364 pagesMarch.
- Boschma, R., Frenken, K., 2011. Technological relatedness and regional branching. In: Bathelt, H., Feldman, M.P., Kogler, D.F. (Eds.), *Dynamic Geographies of Knowledge Creation and Innovation*. Taylor and Francis, Routledge.
- Boschma, R., Minondo, A., Navarro, M., 2012. Related variety and regional growth in Spain. *Papers Regional Sci.* 91 (2).
- Boschma, R., Minondo, A., Navarro, M., 2013. The emergence of new industries at the regional level in Spain: a proximity approach based on product relatedness. *Econ. Geogr.* 89 (1), 29–51.
- Campa, R., 2018. Technological unemployment. a brief history of an idea. *Orbis Idearum. Eu. J. History Ideas* 6 (2).
- Chui, M., Manyika, J., Miremadi, M., 2020. The future of work: how new technologies are transforming tasks. McKinsey Quarterly. Available at: <https://www.mckinsey.com/featured-insights/future-of-work>.
- Cockburn, I.M., Henderson, R., Stern, S., 2019. 4. The impact of artificial intelligence on innovation: an exploratory analysis. In: Agrawal, A., Gans, J., Goldfarb, A. (Eds.), *The Economics of Artificial Intelligence*. University of Chicago Press, pp. 115–148.
- Content, J., Frenken, K., 2016. Related variety and economic development: A literature review. *Eur. Plan. Stud.* 24 (12), 2097–112.
- Damioli, G., Van Roy, V., Vertesy, D., Vivarelli, M., 2021. May AI revolution be labour-friendly? some micro evidence from the supply side. SSRN Electr. J.
- DESA, 2017. The impact of the technological revolution on labour markets and income distribution. *Frontier Issues* (accessed 6 June 2020).
- Dosi, G., Piva, M., Virgillito, M.E., Vivarelli, M., 2019. Embodied and disembodied technological change: the sectoral patterns of job-creation and job-destruction. *Res. Policy.* 50 (4), 104199. May 2021.
- Freire, C., 2017. *Diversification and Structural Economic Dynamics*. PhD dissertation Maastricht University /United Nations University (UNU-MERIT).
- Freire, C., 2019. Economic diversification: a model of structural economic dynamics and endogenous technological change. *Struct. Chang. Econ. Dyn.* 4913–4928.
- Frey, C.B., Osborne, M.A., 2017. The future of employment: how susceptible are jobs to computerisation? *Technol. Forecast. Soc. Change* 114254–114280.
- Füller, J., Hutter, K., Wahl, J., Bilgram, V., Tekic, Z., 2022. How AI revolutionizes innovation management – perceptions and implementation preferences of AI-based innovators. *Technol. Forecast. Soc. Change*, 178121598.
- Furman, J., Seamans, R., 2019. AI and the economy. *Innov. Pol. Econ.* 19 (1), 161–191.
- Graetz, G., Michaels, G., 2018. Robots at work. *Rev. Econ. Stat.* 100 (5), 753–768 (2018).
- Giuntella O., Lu Y., Wang T. (2022) *How do workers and households adjust to robots? Evidence from China*. NBER Working Paper Series. Working Paper 30707.
- Hausmann, R., Klinger, B., 2006. CID Working Paper No. 128.
- Hausmann, R., Klinger, B., 2007. *The Structure of the Product Space and the Evolution of Comparative Advantage*. Harvard University, Center for International Development, Cambridge, MA. CID Working Paper No.146, April.
- Hidalgo, C., Klinger, B., Barabasi, A.-L., Hausmann, R., 2007. The product space conditions the development of nations. *Science* (1979) 317 (5837), 482–487.
- IEEE Spectrum: Technology, Engineering, and Science News (2019). AI and the future of work: the prospects for tomorrow's jobs.
- Kaplan, A., Haenlein, M., 2019. Siri, Siri, in my hand: who's the fairest in the land? ON the interpretations, illustrations, and implications of artificial intelligence. *Bus. Horiz.* 62 (1), 15–25.
- Kauffman, S.A., 2010. *Reinventing the Sacred*. Basic Books, New York.
- Klenert, D., Fernández-Macías, E., Antón, J.I., 2023. Do robots really destroy jobs? evidence from Europe. *Econ. Ind. Democr.* 44 (1), 280–316.
- Mckinsey Global Institute, 2017. A future that works: automation, employment, and productivity. (accessed 19 May 2020). Mitchell, T. M. (2006). the discipline of machine learning. *Mach. Learn.* 66 (1), 2–12.
- Mokyr, J., Vickers, C., Ziebarth, N.L., 2015. The history of technological anxiety and the future of economic growth: is this time different? *J. Econ. Perspect.* 29 (3), 31–50.
- Mühlroth, C., Grottko, M., 2022. Artificial intelligence in innovation: how to spot emerging trends and technologies. *IEEE Trans. Eng. Manage* 69 (2), 493–510.
- Nedelkoska L. and Quintini G. (2018). Automation, skills use and training.
- Neffke F. and Henning M. (2008). *Revealed relatedness: mapping industry space*. Working Paper Series 08.19, Papers in Evolutionary Economic, Utrecht, the Netherlands.
- Neffke, F., Henning, M., Boschma, R., 2011. How do regions diversify over time? Industry relatedness and the development of new growth paths in regions. *Econ. Geogr.* 87 (3), 237–265 (July 2011).
- Pasinetti, 1993. *Structural Economic Dynamics*. Cambridge University Press, Cambridge.
- Pasinetti, L., 1993. *Structural Economic Dynamics*. Cambridge University Press, Cambridge England; New York.
- PwC (2018). Will robots really steal our jobs? an international analysis of the potential long term impact of automation. 47.
- Russell, S., 2020. *Human Compatible: Artificial Intelligence and the Problem of Control*. Penguin Books.
- Tegmark, M., 2020. *Life 3.0: Being Human in the Age of Artificial Intelligence*. Penguin Books.
- The New York Times (2020). The robots are coming. prepare for trouble. 30 january.
- UNCTAD, 2017. *Robots and inclusive growth. Trade and Development Report, 2017*. Ch. 3UNCTAD/TDR/2017.
- UNCTAD (2019). Structural transformation, industry 4.0 and inequality: science, technology and innovation policy challenges. TD/B/C.II/43.
- UNCTAD (2021). Technology and innovation report 2019, *Catching technological waves: Innovation with equity*. UNCTAD/TIR/2020.
- United Nations (2019). Report of the secretary-general on the impact of rapid technological change on sustainable development. E/CN.16/2019/2. Available at

- <https://documents-dds-ny.un.org/doc/UNDOC/GEN/G19/058/47/pdf/G1905847.pdf?OpenElement> (accessed 6 June 2020).
- Varian H. R. (2021). *Artificial Intelligence, economics, and industrial organization*. National Bureau of Economic Research (NBER) Working Paper No. 28514.
- Vermeulen, B., Kesselhut, J., Pyka, A., Saviotti, P., 2018. The impact of automation on employment: just the usual structural change? *Sustainability*. 10 (5), 1661.
- Vivarelli, M., 2014. Innovation, employment and skills in advanced and developing countries: a survey of economic literature. *J. Econ. Issues*. 48 (1), 123–154.
- West, D.M., 2020. *The Future of Work: Robots, AI, and Automation*. Brookings Institution Press.
- Zuboff, S., 2020. *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*. Public Affairs books.